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WordPub

BECE

Integrated Science

Past Questions & Answers

2014 - 2017

**Junior High School
Years 1, 2 & 3**

Compiled by:



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ACKNOWLEDGEMENTS

- The **West African Examination Council** - for the BECE Integrated Science past questions used in this document.
- **Ernest Ato Bentil and Francisco Ameyedowo** – for your invaluable contribution to this work.
- **Ruth, Baaba and Fiifi Arthur** – for your relentless support
- **Teachers, parents and educational institutions** – for your continuous patronage and tireless efforts at ensuring quality education for the 21st century child
- Junior High School **students** – you are the ultimate reason for this work.

DEDICATION

To the Lord **Jesus Christ**, our Saviour and soon-coming King

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JUNE 2017

INTEGRATED SCIENCE 1

OBJECTIVE TEST

45 minutes

*Answer **all** the questions*

Do all rough work on this question paper

1. The instrument used to measure relative humidity is the
 - A. hydrometer.
 - B. hygrometer.
 - C. rain gauge.
 - D. wind vane.

2. The S.I. unit for density is
 - A. ms^{-1}
 - B. ms^{-2}
 - C. kg m^{-3}
 - D. $\text{m}^3 \text{kg}^{-1}$

3. When light travels from glass to air, its speed is
 - A. increased.
 - B. reduced.
 - C. halved.
 - D. unchanged.

4. When a piece of iron rod is brought near a permanent magnet for a while it becomes magnetised by
 - A. stroking method.
 - B. electrical method.
 - C. friction.
 - D. induction.

5. A rheostat is used in an electric circuit in order to
 - A. increase electric current only.
 - B. decrease electric current only.
 - C. keep electric current constant
 - D. increase and decrease electric current.

6. Which of the following statements about an opaque object is **correct**? An opaque object
 - A. allows light to pass through it partially.
 - B. allows light to pass through if fully.
 - C. forms a shadow.
 - D. does not form a shadow

7. One of the characteristics of a solid is that
- A. its particles are tightly packed.
 - B. it assumes the shape of the container it occupies.
 - C. it has no fixed volume.
 - D. its particle can easily be separated.
8. The instrument used to measure current in an electrical circuit is the
- A. ammeter.
 - B. photometer.
 - C. thermometer.
 - D. voltmeter.
9. The bottom of a pool of water appears closer to the surface than it is. Which of the following properties accounts for this observation?
- A. Reflection
 - B. Transparency of water
 - C. Refraction
 - D. Rectilinear propagation
10. An example of a source of energy which is non-renewable is
- A. petroleum.
 - B. solar.
 - C. tide.
 - D. wind.
11. Which of the following apparatus is **not** required in the process of distillation in the laboratory?
- A. Condenser
 - B. Evaporating dish
 - C. Bunsen burner
 - D. Round bottom flask
12. The number of oxygen atoms present in three molecules of carbon dioxide is
- A. 3.
 - B. 4.
 - C. 5.
 - D. 6.
13. The chemical formula Cl_2 represents two
- A. atoms of chlorine.
 - B. neutrons of chlorine.
 - C. ions of chlorine.
 - D. molecules of chlorine.
14. Which of the following substances can rust?
- A. Aluminium
 - B. Bauxite

- C. Copper
- D. Steel

15. Energy stored in food is referred to as

- A. chemical energy.
- B. heat energy.
- C. solar energy.
- D. kinetic energy.

16. Which of the following substances is a mixture?

- A. Water
- B. Sodium chloride
- C. Sodium chloride solution
- D. Iron filing

17. The mole is the S.I. unit of

- A. amount of substance.
- B. luminous intensity.
- C. temperature.
- D. mass of substance.

18. An atom has a proton number of 17 and neutron number 18. Determine the number of electrons in the third shell of the atom.

- A. 1
- B. 3
- C. 5
- D. 7

19. An example of mixtures that can be separated by the method of filtration is

- A. sugar in water.
- B. sand in water.
- C. oil in water.
- D. ink in water.

20. Which of the following liquids would turn blue litmus paper red?

- A. Vinegar
- B. Water
- C. Ammonia solution
- D. Bicarbonate solution

21. Which of the following particles constitute matter?

- I. Atoms
- II. Molecules
- III. Ions

- A. I only
- B. II only

- C. II and III only
- D. I, II and III

22. An example of inheritable characteristics in humans is
- A. shape of nose.
 - B. knowledge.
 - C. ability to speak different languages.
 - D. handwriting.
23. Which of the following statements about aerobic respiration is **correct**?
- A. The by-products are carbon dioxide and water.
 - B. Little amount of energy is produced.
 - C. Oxygen is not required.
 - D. The by-products are alcohol and carbon dioxide.
24. Sperms produced by the testes in humans are temporarily stored in the
- A. epididymis.
 - B. prostate gland.
 - C. seminal vesicle.
 - D. sperm duct.
25. Which of the following changes normally occur during old age in humans?
- A. Grey hair starts appearing
 - B. Mental alertness increases
 - C. Broadening of the chest
 - D. Attraction to the opposite sex
26. One of the characteristics of wind pollinated flowers is that they
- A. are large and conspicuous.
 - B. are usually scented.
 - C. have nectar.
 - D. have feathery stigma.
27. The testes is normally located outside the body of the male because
- A. the temperature of the body is too high for effective sperm production.
 - B. it cannot increase in size when it is in the body.
 - C. it can absorb moisture from the atmosphere.
 - D. it occupies too much space.
28. The organelle which occupies the **largest** portion of a plant cell is the
- A. chloroplast.
 - B. mitochondrion.
 - C. nucleus.
 - D. vacuole.
29. Movement of the ribs is brought about by the
- A. pleural cavity.

- B. intercostal muscle.
- C. larynx.
- D. trachea.

30. Fertilisation in humans usually occurs in the
- A. ovary.
 - B. uterus.
 - C. vagina.
 - D. fallopian tube.
31. Which of the following processes is an example of osmosis?
- A. Spreading of perfume in a room
 - B. Absorption of food nutrients into the small intestines
 - C. Spreading of potassium permanganate in water
 - D. Absorption of water into the root hairs
32. The part of a living cell which is semi-permeable is
- A. cytoplasm.
 - B. nucleus.
 - C. protoplasm.
 - D. cell membrane.
33. Which of the following statements about soil profile is/are **correct**? It helps the farmer to determine
- I. soil fertility.
 - II. water-holding capacity of the soil.
 - III. depth of the soil.
- A. I only
 - B. I and II only
 - C. II and III only
 - D. I, II and III
34. Which of the following practices is **not** a principle of crop rotation?
- A. Legumes should be added to the rotation programme.
 - B. Deep rooted crops should be followed by shallow rooted crops.
 - C. Crops that are closely related should follow each other.
 - D. Fallow periods should be allowed during the time of rotation.
35. The relative proportion of sand, silt and clay in a given sample of soil is soil
- A. profile.
 - B. porosity.
 - C. structure.
 - D. texture.
36. The **first** factor to be considered when deciding to cultivate a particular vegetable is
- A. cultural practices.
 - B. selection of land.

- C. harvesting.
D. method of propagation.
37. Which of the following factors should be considered when selecting a site for vegetable crop production?
- I. Nearness to the market.
II. Topography of the land.
III. Nearness to the sea
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
38. Which of the following activities are cultural practices in the cultivation of cabbage?
- I. Pruning
II. Weeding
III. Watering
- A. I and II only
B. I and III only
C. II and III only
D. I, II and III
39. An example of a vegetable crop is
- A. cocoa.
B. cotton.
C. potato.
D. tomato.
40. The majority charge carriers in a p-type semi-conductor are
- A. electrons.
B. holes.
C. neutrons.
D. protons.

END OF PAPER

JUNE 2017

INTEGRATED SCIENCE 1

OBJECTIVE TEST SOLUTIONS

1. B. hygrometer.
2. C. kg m^{-3}
3. A. increased
4. D. induction
5. D. increase and decrease electric current
6. C. forms a shadow
7. A. its particles are tightly packed
8. A. ammeter
9. C. Refraction
10. A. petroleum
11. B. Evaporating dish
12. D. 6
13. A. atoms of chlorine
14. D. Steel
15. A. chemical energy
16. C. Sodium chloride solution
17. A. amount of substance
18. D. 7
19. B. sand in water
20. A. Vinegar
21. D. I, II and III
22. A. shape of nose
23. A. The by-products are carbon dioxide and water
24. A. epididymis
25. A. Grey hair starts appearing
26. D. have feathery stigma
27. A. the temperature of the body is too high for effective sperm production
28. D. vacuole
29. B. intercostal muscle
30. D. fallopian tube

- 31. D. Absorption of water into the root hairs
- 32. D. cell membrane
- 33. D. I, II and III
- 34. C. Crops that are closely related should follow each other
- 35. D. texture
- 36. B. selection of land
- 37. A. I and II only
- 38. D. I, II and III
- 39. C. potato
- 40. B. holes

JUNE 2017

INTEGRATED SCIENCE 2

ESSAY [100 marks]

1 ¼ hours

*This paper is in **two** sections: **A** and **B**. Answer Question 1 in section **A** and any other **four** questions in section **B**. Credit will be given for clarity of expression and orderly presentation of material.*

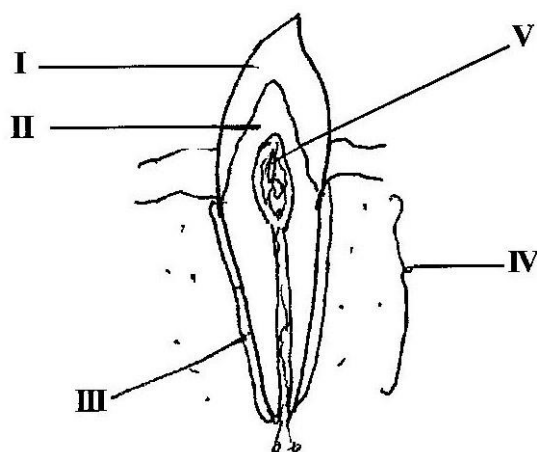
SECTION A

[40 marks]

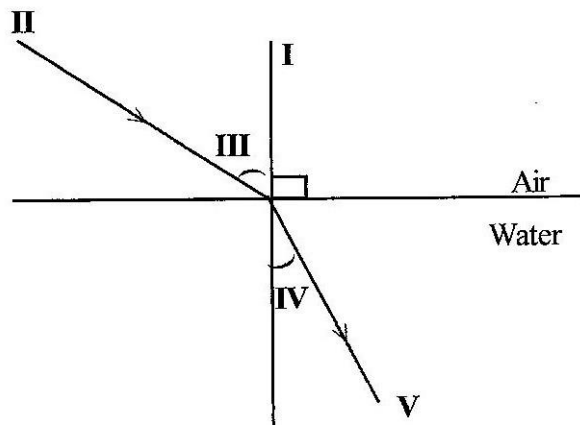
Answer **all** of Question 1

1. (a) The diagram below is an illustration of a longitudinal section of a canine tooth in humans.

Study the diagram carefully and answer the questions that follow.

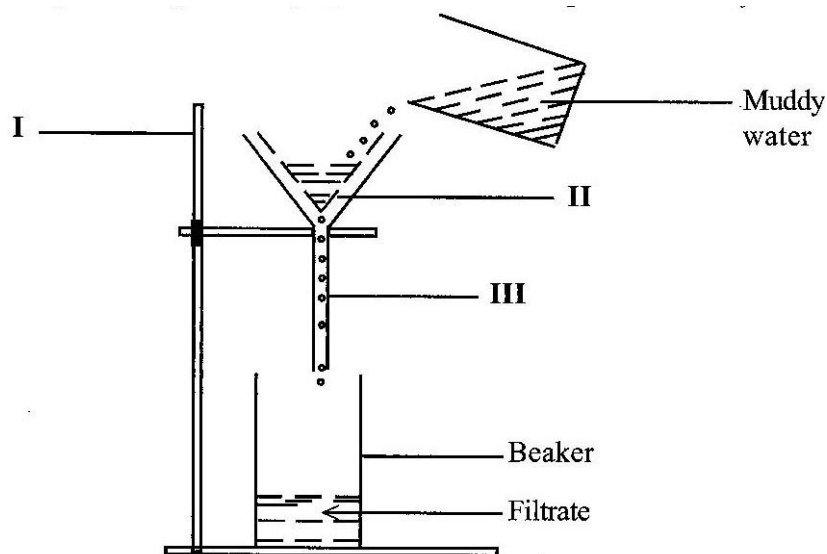


- (i) Name **each** of the parts labelled **I, II, III, IV** and **V**. [5 marks]
(ii) What is the function of **each** of the parts labelled **I** and **III**? [2 marks]
(iii) Which of the labelled parts could be affected by tooth decay? [1 mark]
(iv) State **three** ways by which tooth decay may be prevented. [3 marks]
- (b) The diagram below is an illustration of a scientific phenomenon which occurs in nature.
Study the diagram carefully and answer the questions that follow.



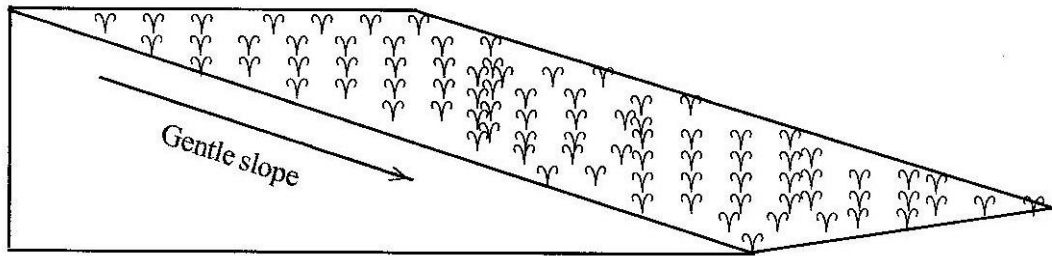
- (i) What phenomenon does the diagram illustrate? [1 mark]
- (ii) Identify **each** of the parts labelled **I, II, III, IV** and **V**. [5 marks]
- (iii) Explain why an object at the bottom of a pond appears closer to the surface than it actually is. [3 marks]
- (c) The diagram below is an illustration of an experiment performed to separate the components of muddy water.

Study the diagram carefully and answer the questions that follow.



- (i) Name **each** of the parts labelled **I, II** and **III**. [3 marks]
- (ii) State the function of the part labelled **II**. [1 mark]
- (iii) Name the substance obtained as the filtrate. [1 mark]
- (iv) State **three** physical properties of the filtrate. [3 marks]
- (v) Name **two** other materials that could be used in place of the part labelled **II**. [2 marks]
- (d) The diagram below illustrates a farmland on a slopy area.

Study the diagram carefully and answer the questions that follow.



- (i) What process is **likely** to occur on the farmland when it rains heavily? [1 mark]
- (ii) State **two** farming practices that can also lead to the process mentioned in (i). [2 marks]
- (iii) List **four** farming practices that could be used to control the process mentioned in (i). [4 marks]
- (iv) Mention **three** soil resources that would be depleted from the farmland when it rains heavily. [3 marks]

SECTION B
[60 marks]

*Answer **four** questions **only** from this section*

2. (a) An atom Y has atomic number **12**. It loses two electrons in order to be stable.
- (i) State the proton number of the atom before it loses electrons. [1 mark]
- (ii) State the electron number of the atom:
- (α) before it loses electrons.
- (β) after losing electrons. [2 marks]
- (iii) Name the type of ion formed by the atom when it loses **two** electrons. [1 mark]
- (b) Name **four** farming systems used in crop production. [4 marks]
- (c) (i) What is *dispersal of seeds*? [2 marks]
- (ii) State **two** characteristics of seeds dispersed by wind. [2 marks]
- (d) Explain the term *forward bias* of a p-n junction diode. [3 marks]
3. (a) (i) What is an acid? [2 marks]
- (ii) Give **two** differences between an *acid* and a *base*, in terms of taste and feel. [2 marks]
- (b) (i) Define *pressure*.
- (ii) A force of 200 N is exerted on an area of 50 m². Calculate the pressure exerted by the force. [3 marks]
- (c) Explain the following terms as associated with living organisms:
- (i) unicellular;
- (ii) multicellular. [4 marks]
- (d) Give **two** reasons why soil air is important. [2 marks]
4. (a) (i) Explain the following terms as applied to machines:
- (α) work input;
- (β) work output. [4 marks]
- (ii) State **one** factor that limits work output for a given work input in a simple machine. [1 marks]

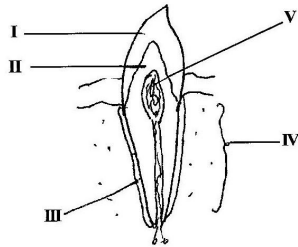
- (b) (i) What is *chloroplast*? [2 marks]
(ii) Differentiate between *aerobic respiration* and *anaerobic respiration*. [2 marks]
- (c) (i) State the colour change that would occur when blue litmus paper is dipped into a solution of:
(α) vinegar;
(β) wood ash. [2 marks]
(ii) Name the products formed when hydrochloric acid reacts with sodium hydroxide. [2 marks]
- (d) List **two** benefits of vegetables to humans. [2 marks]
5. (a) (i) Differentiate between *egestion* and *digestion* in nutrition. [2 marks]
(ii) What is the end-product of digestion? [2 marks]
- (b) Give **one** example of a chemical compound used in:
(i) medicine;
(ii) agriculture;
(iii) industry. [3 marks]
- (c) (i) Define the term *soil profile*. [2 marks]
(ii) State **two** ways in which soil profile is important in crop production. [2 marks]
- (d) (i) State **two** steps used by scientists in doing their work. [2 marks]
(ii) Give **two** subjects that may be considered as applied sciences. [2 marks]
6. (a) (i) What is an *alloy*? [2 marks]
(ii) State **two** causes of corrosion of metals. [2 marks]
- (b) (i) What is a *planet*? [2 marks]
(ii) Name **two** planets between the Sun and the Earth. [2 marks]
- (c) State **four** functions of the circulatory system in humans. [4 marks]
- (d) (i) Define the term *crop rotation*. [2 marks]
(ii) Give **one** example of a chemical method of controlling pests on crop farms. [marks]

END OF ESSAY TEST

ESSAY

SOLUTIONS

1. (a) The diagram below is an illustration of a longitudinal section of a canine tooth in humans. Study the diagram carefully and answer the questions that follow.



- (i) Name each of the parts labelled I, II, III, IV and V. [5 marks]

I	-	Crown / Enamel
II	-	Dentine
III	-	Cement / Periodontal membrane
IV	-	Jaw bone / Root
V	-	Pulp Cavity

- (ii) What is the function of each of the parts labelled I and III? [2 marks]

I	-	Protects teeth from harmful bacteria
	-	Enable teeth withstand pressure of chewing
	-	Cutting / chewing / protects dentine / protect pulp cavity
III	-	hold tooth firmly (in socket / Jaw bone)

- (iii) Which of the labelled parts could be affected by tooth decay? [1 mark]

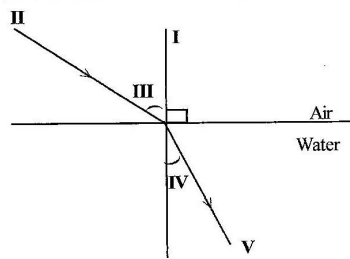
V	-	Pulp cavity
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- (iv) State three ways by which tooth decay may be prevented. [3 marks]

- avoid using sharp-pointed objects
- avoid having dry mouth
- use water containing fluorides to wash teeth / mouth
- avoid smoking
- the teeth should be cleaned regularly after each meal / remove food particle from the teeth / flossing
- wash the mouth vigorously after each meal
- regular visit to the dentist

- avoid eating too hot foods
- avoid eating too much sugary food
- avoid eating too cold food / taking too cold drinks

- (b) The diagram below is an illustration of a scientific phenomenon which occurs in nature. Study the diagram carefully and answer the questions that follow.



- (i) What phenomenon does the diagram illustrate? [1 mark]

Refraction of light

- (ii) Identify each of the parts labelled I, II, III, IV and V. [5 marks]

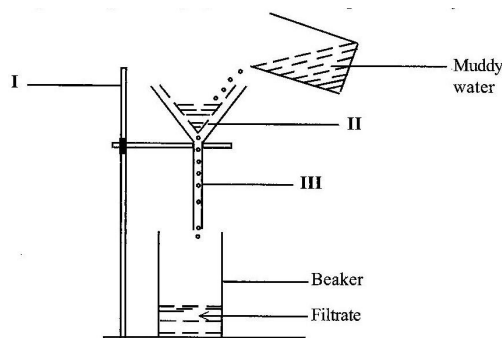
- I - Normal
- II - Incident ray
- III - angle of incidence / incident angle
- IV - angle of refraction
- V - refracted ray

- (iii) Explain why an object at the bottom of a pond appears closer to the surface than it actually is. [3 marks]

Rays from the object at the bottom are bent away from the normal as they come out of the water. In a straight line, these rays appear to come from a point above the bottom

- (c) The diagram below is an illustration of an experiment performed to separate the components of muddy water.

Study the diagram carefully and answer the questions that follow.



(i) Name each of the parts labelled I, II and III. [3 marks]

- I - Retort / clamp stand
- II - filter paper
- III - funnel

(ii) State the function of the part labelled II. [1 mark]

- to prevent the residue / solid / insoluble particles / mud from entering the filtrate
- to filter (solid / insoluble particles / residue)
- to separate the mud, solid / insoluble particles from the water

(iii) Name the substance obtained as the filtrate. [1 mark]

Water

(iv) State three physical properties of the filtrate. [3 marks]

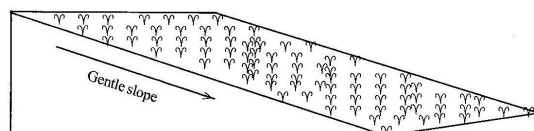
- it has a density of $1 \text{ g cm}^{-3} / 1000 \text{ kg m}^{-3}$
- it boils at 100°C
- it freezes at 0°C (at 1 atm)
- it is colourless
- it is tasteless / insipid
- it is odourless
- it has a high surface tension, etc

(v) Name two other materials that could be used in place of the part labelled II. [2 marks]

- foam
- cotton wool
- glass wool
- Clean / white cloth

(d) The diagram below illustrates a farmland on a sloppy area.

Study the diagram carefully and answer the questions that follow.



(i) What process is likely to occur on the farmland when it rains heavily? [1 mark]

Erosion

(ii) State two farming practices that can also lead to the process mentioned in (i). [2 marks]

- Deforestation
- bush burning
- continuous cropping
- cultivation / ploughing along slope
- crop removal
- excessive use of chemical / inorganic fertilizers
- overgrazing

(iii) List four farming practices that could be used to control the process mentioned in (i). [4 marks]

- planting cover crops
- terracing
- cultivating / ploughing across slope / contour
- heavy mulching
- constructing drainage channels
- planting trees beside farm
- strip cropping
- contour bunding

(iv) Mention three soil resources that would be depleted from the farmland when it rains heavily. [3 marks]

- soil organisms
- nutrients (macro / micro) / mineral matter
- soil particles / mineral matter
- growing crops
- air
- organic matter / humus

SECTION B

[60 marks]

Answer four questions only from this section

2. (a) An atom Y has atomic number 12. It loses two electrons in order to be stable.

(i) State the proton number of the atom before it loses electrons. [1 mark]

12

(ii) State the electron number of the atom: [2 marks]

(α) before it loses electrons.

(β) after losing electrons.

(iii) Name the type of ion formed by the atom when it loses two electrons. [1 mark]

Cation / positively charged ion

(b) Name four farming systems used in crop production. [4 marks]

- Land rotation
- Crop rotation
- Mixed cropping
- Mixed farming
- Organic farming
- Monoculture
- Mono cropping
- Shifting cultivation
- Ecological farming / Eco-farming

(c) (i) What is *dispersal of seeds*? [2 marks]

It is a process by which seeds are carried away from the parent plant

(ii) State two characteristics of seeds dispersed by wind. [2 marks]

- seeds are very small
- they are very light
- they have hair
- they have wing-like structures / parachutes

(d) Explain the term *forward bias* of a p-n junction diode. [3 marks]

It is when the positive terminal of an electric source is connected to the p-type of the diode and the negative terminal of the source is connected to the n-type of the diode, resulting in the flow of charges / current.

3. (a) (i) What is an acid? [2 marks]

It is a proton donor

OR

A substance or compound that produces excess hydrogen ion (H⁺) in water

OR

A substance that contains replaceable hydrogen

- (ii) Give two differences between an *acid* and a *base*, in terms of taste and feel. [2 marks]

	Acid	Base
Taste	Sour	Bitter
Feel	Non slippery / stinging	Slippery / soapy

- (b) (i) Define *pressure*.

It is the force acting (normally) per unit area
OR

$$\text{Pressure} = \frac{(\text{Normal})\text{force}}{\text{Area}}$$

- (ii) A force of 200 N is exerted on an area of 50 m². Calculate the pressure exerted by the force. [3 marks]

$$\begin{aligned}\text{Pressure} &= \frac{(\text{Normal})\text{force}}{\text{Area}} \\ &= \frac{200}{50} \\ &= 4 \text{ Pa} \quad \text{OR} \quad 4 \text{ Nm}^{-2}\end{aligned}$$

- (c) Explain the following terms as associated with living organisms: [4 marks]

- (i) **Unicellular** -
(Very small) living organisms that consist of only one cell.
- (ii) **Multicellular** -
(Large) living organisms that consist of (many) cells

- (d) Give two reasons why soil air is important. [2 marks]

- It enhances the absorption of mineral salts / nutrients by plant roots
- It enhances the absorption of water by plant roots
- It makes oxygen available for seed germination
- Presence of aeration prevents the formation of toxic / acidic substances in the soil by micro organisms.
- Plant roots use soil air for respiration / metabolism / growth
- Soil micro organisms use soil air for respiration
- Air is required for the decomposition of organic matter.
- Aeration is required to prevent development of plant diseases

4. (a) (i) Explain the following terms as applied to machines: [4 marks]
(α) work input;

It is total energy / work applied to a machine for it to be able to work

- (β) work output.

It is the total energy / work obtained by using machines to do work

- (ii) State one factor that limits work output for a given work input in a simple machine. [1 marks]

- friction / wear and tear
- age of machine / period of usage
- weight of machine parts

- (b) (i) What is *chloroplast*? [2 marks]

It is an organelle / structure found only in (plant) cell and it contains chlorophyll / green pigment

- (ii) Differentiate between *aerobic respiration* and *anaerobic respiration*. [2 marks]

Aerobic Respiration	Anaerobic Respiration
Requires oxygen	Does not require oxygen
Produces large amount of energy	Produces small amount of energy
Water is produced	Ethanol / lactic acid is produced

- (c) (i) State the colour change that would occur when blue litmus paper is dipped into a solution of: [2 marks]

- (α) vinegar; - changes to red
(β) wood ash. - there is no colour change / remains the same

- (ii) Name the products formed when hydrochloric acid reacts with sodium hydroxide. [2 marks]

Sodium chloride and water

- (d) List two benefits of vegetables to humans. [2 marks]

- Prevents constipation
- supply roughage for easy digestion of food.
- to garnish / decorate food

- provide humans with vitamins / proteins / mineral salts / carbohydrates / food nutrients / food
- source of medicine

5. (a) (i) **Differentiate between *egestion* and *digestion* in nutrition.** [2 marks]

Egestion is the removal of undigested food / semi-solid / waste / faeces from the anus while digestion is the breaking down of food into smaller building blocks / components that can be absorbed into the blood stream.

(ii) **What is the end-product of digestion?** [2 marks]

- glucose / simple sugar
- amino acids
- fatty acids and glycerol

(b) **Give one example of a chemical compound used in:** [3 marks]

(i) medicine;

Drugs / other pharmaceutical products, etc

(ii) agriculture;

Insecticides, pesticides, fertilizers, etc

(iii) industry.

Hydrochloric acid, sodium chloride, ethanol, ammonia, nitric acid, etc

(c) (i) **Define the term *soil profile*.** [2 marks]

It is the vertical section of the land showing the various horizons / layers and their composition

(ii) **State two ways in which soil profile is important in crop production.** [2 marks]

- helps to know the type of tools to choose
- helps to select good soil for the crop
- to determine depth at which tillage implements must be set
- to know soil management practices to adopt
- helps to determine fertility of soil
- helps to determine type of crop to grow
- helps to know the water holding capacity of the soil.
- helps to determine the type of fertilizer to use / apply

(d) (i) **State two steps used by scientists in doing their work.** [2 marks]

- Identifying the problem
- Observation
- Experimenting
- Analyzing data
- Drawing conclusions
- Hypothesis
- Collection of data

(ii) Give two subjects that may be considered as applied sciences. [2 marks]

- Engineering
- Medicine
- Agriculture
- Pharmacy
- Electronics
- ICT
- etc

6. (a) (i) What is an *alloy*? [2 marks]

It is a uniform mixture of a metal and another metal or non-metal or two or more metals

(ii) State two causes of corrosion of metals. [2 marks]

- Presence of oxygen / air
- Presence of moisture / water
- Presence of alkali
- Presence of salt
- Presence of acid

(b) (i) What is a *planet*? [2 marks]

It is a heavenly body that moves round the sun / star

(ii) Name two planets between the Sun and the Earth. [2 marks]

- Mercury
- Venus

(c) State four functions of the circulatory system in humans. [4 marks]

- Transports blood
- Transports hormones / drugs / chemical
- Transports oxygen to cells / tissues / organs of the body
- Transports nutrients / glucose to organs

- Carries away excretory products / CO₂ / waste materials / urea.
- Regulates body temperatures / transport
- Consists of / transports white blood cells / antibodies / attack foreign bodies.
- Transports blood platelets to clot blood
- responsible for pumping blood / erection of penis / clitoris

(d) (i) Define the term *crop rotation*.

[2 marks]

It is a system of farming where different types of crops are grown on the same piece of land but on different plots in a definite order / cycle / sequence from season to season.

OR

It is a system of farming where variety of crops are grown on the same piece of land in a repeated cycle

(ii) Give one example of a chemical method of controlling pests on crop farms.

[1 mark]

- spraying of recommended / appropriate pesticides
- placement of recommended / appropriate pesticides

JUNE 2016

INTEGRATED SCIENCE 1

OBJECTIVE TEST

45 minutes

1. An example of a third class lever is
 - A. wheelbarrow
 - B. sugar tongs
 - C. pair of scissors
 - D. crowbar
2. The release of a mature egg from an ovary into the fallopian tube in humans is called
 - A. copulation
 - B. ejaculation
 - C. menstruation
 - D. ovulation
3. The blood vessel that carries oxygenated blood from the lungs to the heart is known as
 - A. pulmonary artery
 - B. pulmonary vein
 - C. vena cava
 - D. aorta
4. An example of a plant micro-nutrient is
 - A. calcium
 - B. copper
 - C. magnesium
 - D. potassium
5. What is the colour of the neutral wire in a three-pin plug?
 - A. Blue
 - B. Brown
 - C. Green
 - D. Yellow
6. In the pin-hole camera, the image formed is always
 - A. erect and bright
 - B. erect and blurred
 - C. inverted and real
 - D. inverted and virtual
7. The part of the flower that develops into a fruit is the
 - A. ovary

- B. ovule
- C. stamen
- D. style

8. The number of elements in the compound $\text{Ca}(\text{OH})_2$ is

- A. 2
- B. 3
- C. 4
- D. 5

9. The reason why alum is added to water during treatment is to

- A. kill germs
- B. give taste to water
- C. make water colourless
- D. make suspended particles settle

10. Which of the following animal parasites could be controlled by hand picking?

- A. Liver fluke
- B. Tapeworm
- C. Tick
- D. Roundworm

11. A stick which is partially immersed in water appeared to be bent due to

- A. absorption
- B. reflection
- C. refraction
- D. transmission

12. One difference between metals and non-metals is that metals

- A. have low density
- B. are not malleable
- C. have luster
- D. have low melting points.

13. The cultivation of different crops on different plots of a farmland in a definite cycle is called

- A. land rotation
- B. shifting cultivation
- C. crop rotation
- D. mixed cropping

14. Iodine deficiency in humans could result in a disorder known as

- A. diabetes
- B. goiter
- C. kwashiorkor
- D. scurvy

15. A substance is termed combustible if it

- A. easily catches fire
 - B. dissolves common salt
 - C. sublimates at room temperature
 - D. boils at 100°C.
16. Micro-organisms that cause diseases are collectively called
- A. bacteria
 - B. infections
 - C. pathogens
 - D. viruses
17. Which type of energy is lost when sweat evaporates from the human body?
- A. Sound energy
 - B. Mechanical energy
 - C. Chemical energy
 - D. Heat energy
18. Which of the following gases is involved in the rusting of iron?
- A. Hydrogen
 - B. Oxygen
 - C. Nitrogen
 - D. Carbon dioxide
19. The feeling of soil between fingers is used to determine the
- A. texture of the soil
 - B. drainage of the soil
 - C. capillarity of the soil
 - D. water holding capacity of the soil.
20. Sodium hydroxide is an example of a base because it
- A. has sour taste
 - B. has a pH less than 7
 - C. turns wet blue litmus paper red
 - D. turns wet red litmus paper blue
21. Which of the following plants has its leaves modified for storing food?
- A. Ginger
 - B. Onion
 - C. Pineapple
 - D. Tomato
22. Gases enter and leave the leaf of a plant through the
- A. mesophyll cells
 - B. stomata
 - C. pith
 - D. palisade cells

23. Which of the following vegetable crops is cultivated for its leaves?
- A. Carrot
 - B. Lettuce
 - C. Okro
 - D. Pepper
24. An example of a body that does not produce its own light is the
- A. moon
 - B. sun
 - C. star
 - D. fire fly
25. Which of the following devices converts electrical energy into sound energy?
- A. Electric heater
 - B. Electric fan
 - C. Washing machine
 - D. Loudspeaker
26. Transplanting of young seedlings is usually done in the evening because
- A. darkness promotes rapid growth
 - B. seedlings require less nutrients
 - C. pest attack is minimal
 - D. transpiration is minimal
27. A mixture of sugar and water could be separated by
- A. decantation
 - B. evaporation
 - C. filtration
 - D. sublimation
28. Which of the following processes results in the formation of new substances?
- A. Cooling water to form ice
 - B. Adding saliva to cooked yam
 - C. Dissolving sugar in water
 - D. Mixing iron dust and sand
29. Photosynthesis is important to living organisms because it produces
- A. glucose and oxygen
 - B. glucose and carbon dioxide
 - C. oxygen and carbon dioxide
 - D. water and carbon dioxide
30. Global warming is caused by the
- A. circulation of oxygen in the atmosphere
 - B. excessive release of carbon dioxide into the atmosphere
 - C. release of hydrogen into the atmosphere
 - D. circulation of nitrogen in the atmosphere

31. The warning and safety signs on chemical containers are usually represented by a symbol placed within a
- A. circle
 - B. rectangle
 - C. square
 - D. triangle
32. In which of the following structures in a living cell is cellulose found?
- A. Chloroplast
 - B. Mitochondrion
 - C. Cell membrane
 - D. Cell wall
33. Elements that could easily lose electrons to form cations are
- A. metals
 - B. non-metals
 - C. semi-metals
 - D. noble gases
34. The conversion of agricultural produce from its original form to other desirable forms is termed
- A. preservation
 - B. processing
 - C. recycling
 - D. storage
35. The function of the hair in the nose of humans is to
- A. make breathing easier
 - B. keep nose moist
 - C. filter dust from the air breathed in
 - D. keep the nose warm
36. A metal that is used as a thermometric liquid is
- A. aluminium
 - B. copper
 - C. mercury
 - D. silver
37. The joule is the S.I unit for
- A. energy
 - B. force
 - C. power
 - D. pressure
38. Which of the following gases supports combustion?
- A. Hydrogen
 - B. Oxygen

- C. Carbon dioxide
- D. Carbon monoxide

39. The form of energy which flows from one point to another due to temperature differences is known as

- A. mechanical energy
- B. heat energy
- C. solar energy
- D. nuclear energy

40. Which of the following resources does not produce energy?

- A. Coal
- B. Sand
- C. Water
- D. Wind

JUNE 2016

INTEGRATED SCIENCE 1

OBJECTIVE TEST

SOLUTIONS

1. B. sugar tongs
2. D. ovulation
3. B. pulmonary vein
4. B. copper
5. A. Blue
6. C. inverted and real
7. A. ovary
8. B. 3
9. D. make suspended particles settle
10. C. Tick
11. C. refraction
12. C. have luster
13. C. crop rotation
14. B. goiter
15. A. easily catches fire
16. C. pathogens
17. D. Heat energy
18. B. Oxygen
19. A. texture of the soil
20. D. turns wet red litmus paper blue
21. B. Onion
22. B. stomata
23. B. Lettuce
24. A. moon
25. D. Loudspeaker
26. D. transpiration is minimal
27. B. evaporation
28. B. Adding saliva to cooked yam
29. A. glucose and oxygen

- 30. B. excessive release of carbon dioxide into the atmosphere
- 31. D. triangle
- 32. D. Cell wall
- 33. A. metals
- 34. B. processing
- 35. C. filter dust from the air breathed in
- 36. C. mercury
- 37. A. energy
- 38. B. Oxygen
- 39. B. heat energy
- 40. B. Sand

JUNE 2016

INTEGRATED SCIENCE 2

ESSAY

1 ¼ hours

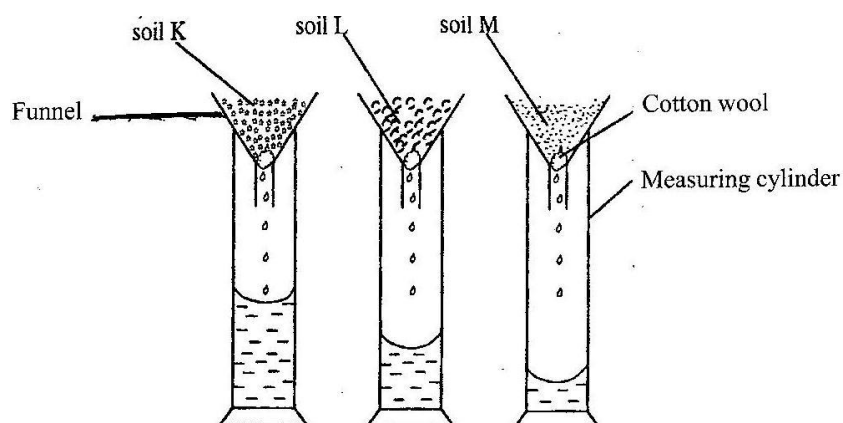
This paper is in two parts: I and II. Answer Question 1 in part I and any other four questions in part II

PART I **[40 marks]**

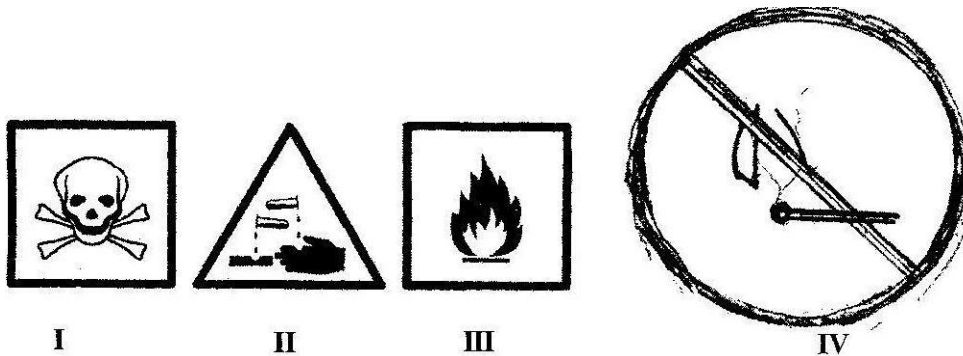
Answer **all** of Question 1

1. (a) The funnels in the diagram below contain equal amounts of different types of soils labelled **K**, **L** and **M**. Equal volumes of water were poured onto each soil at the same time and allowed to drain for 20 minutes.

Study the diagram and answer the questions that follow



- | | | |
|-------|---|-----------|
| (i) | What is the aim of the experiment? | [2 marks] |
| (ii) | Which soil has the highest rate of drainage? | [1 mark] |
| (iii) | Which soil has the highest water retention capacity? | [1 mark] |
| (iv) | Which soil is most likely to lose water and dry faster after rainfall? | [1 mark] |
| (v) | Which soil is most likely to be waterlogged after rainfall? | [1 mark] |
| (vi) | Which of the soil types would be suitable for maize cultivation? | [1 mark] |
- (b) The diagram below illustrates hazard symbols labelled **I**, **II**, **III** and **IV**.
Study the diagram carefully and answer the questions that follow.



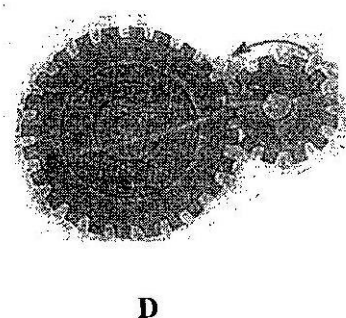
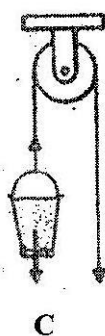
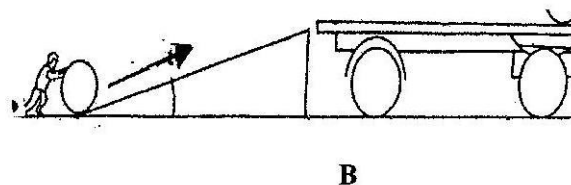
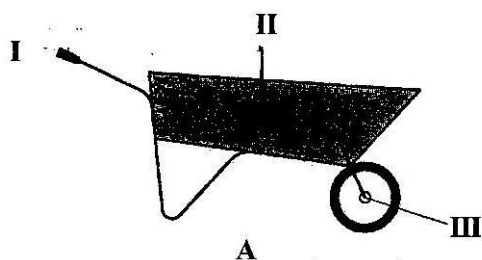
- (i) What does **each** of the symbols labelled **I**, **II**, **III** and **IV** represent?
- (ii) Name **one** substance **each** that is associated with:
 - (α) **I**;
 - (β) **II**;
 - (γ) **III**.
- (iii) Name a place where the hazard symbol labelled **IV** is often displayed
- (iv) Which of the symbol(s) is / are found on chemical containers?

[3 marks]

[1 mark]

[3 marks]

The diagrams below are illustrations of devices used to do work easily
Study the diagrams and answer the questions that follow



- (i) Give a general name for the devices.
- (ii) Identify **each** of the devices labelled **A**, **B**, **C** and **D**.
- (iii) Name the parts labelled **I**, **II** and **III** of device **A** when it is considered as a lever.
- (iv) What does the arrow represent in the device labelled **B**?
- (v) Name the type of work done with **each** of the devices labelled:

[1 mark]

[4 marks]

[3 marks]

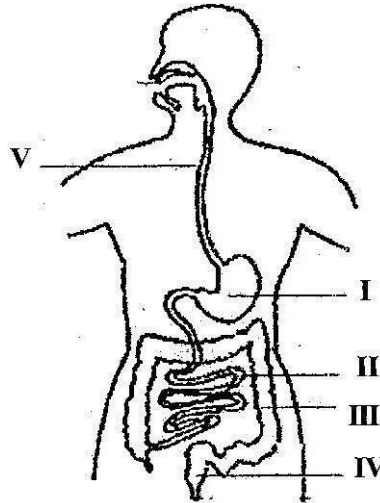
[1 mark]

- (α) **C;**
(β) **D;**

[2 marks]

- (d) The diagram below illustrates the digestive system in humans.

Study the diagram carefully and answer the questions that follow



- (i) Name **each** of the parts labelled **I, II, III, IV** and **V** [5 marks]
- (ii) Name the part(s) of the digestive system where
- (α) digestion of food substances occur
- (β) digested food is absorbed into the bloodstream [3 marks]
- (iii) Name the end-products of the digestion that is absorbed into the bloodstream [3 marks]

PART II
[60 marks]

*Answer **four** questions **only** from this section*

2. (a) Name **four** weather measuring instruments. [4 marks]
- (b) Name the stages in the life cycle of a mosquito. [4 marks]
- (c) (i) List **two** properties of water
(ii) Explain why it is advisable to wash clothes with soft water [4 marks]
- (d) State **three** ways in which soil profile is important. [3 marks]
3. (a) List **three** modes of heat transfer. [3 marks]
- (b) (i) What is a *deficiency disease*?
(ii) Name **three** deficiency diseases in humans. [5 marks]
- (c) State **two** ways in which **each** of the following factors cause depletion of soil resources:
(i) burning
(ii) leaching [4 marks]
- (d) List **three** processes that can change matter from one state to another. [3 marks]
4. (a) (i) What is a *satellite*?
(ii) Give **three** uses of artificial satellites. [5 marks]
- (b) State the composition of **each** of the following alloys:
(i) brass
(ii) steel
(iv) bronze [3 marks]
- (c) Name **three** cultural practices used in vegetable production. [3 marks]
- (d) List **four** parts of the respiratory system in humans. [4 marks]
5. (a) (i) What is a *force*?
(ii) State **two** ways in which forces could affect a body [4 marks]

- (b) (i) What is a *chemical change*?
(ii) State **three** differences between *chemical change* and *physical change*. [5 marks]
- (c) Name **three** physical properties of soil. [3 marks]
- (d) Mention **three** diseases of the circulatory system in humans. [3 marks]

6. (a) Write the systematic name for **each** of the following compounds:
(i) H_2O
(ii) MgO
(iii) CaO
(iv) CaCl_2 [4 marks]
- (b) Name the instrument used in measuring the following:
(i) length of a rope
(ii) mass of a stone
(iii) temperature of a liquid
(iv) volume of a liquid [4 marks]
- (c) List **three** factors that influence vegetable crop production. [3 marks]
- (d) Name **four** stages in the life cycle of a flowering plant. [4 marks]

JUNE 2016

INTEGRATED SCIENCE 2

ESSAY

SOLUTIONS

1. (a) (i) The aim of the experiment [2 marks]
To determine the drainage ability / water-holding capacity of the soils
- (ii) Soil with the highest rate of drainage [1 mark]
Soil K
- (iii) Soil with the highest water retention capacity [1 mark]
Soil M
- (iv) The soil **most** likely to lose water and dry faster after rainfall [1 mark]
Soil K
- (v) The soil **most** likely to be waterlogged after rainfall [1 mark]
Soil M
- (vi) Which of the soil types would be suitable for maize cultivation? [1 mark]
Soil L
- (b) (i) What **each** of the symbols labelled **I, II, III** and **IV** represent
- | | | |
|-----|---|---------------------------------------|
| I | - | Danger |
| II | - | Corrosive |
| III | - | Highly inflammable / highly flammable |
| IV | - | No naked flame |
- (ii) **One** substance **each** that is associated with:
- | | | |
|-----|-------------|--|
| (α) | I; | DDT, Hydrogen cyanide, Salicylic acid |
| (β) | II; | Concentrated Inorganic acids, such as HCl, H ₂ SO ₄ , HNO ₃ ,
Concentrated inorganic bases, such as NaOH, KOH, Ca(OH) ₂
Household bleach |
| (γ) | III. | Petrol, Kerosene, LPG, Perfume, Insecticides, Alcohol |
- [3 marks]
- (iii) A place where the hazard symbol labelled **IV** is often displayed [1 mark]
Gas Filling stations, Storage places of combustible substances

- (iv) Symbol(s) found on chemical containers [3 marks]
I, II and III
- (c) (i) General name for the devices. [1 mark]
Simple machines
- (ii) Identification of **each** of the devices labelled **A, B, C** and **D**. [4 marks]
 A - Wheel barrow
 B - Inclined plane
 C - Pulley
 D - Gear
- (iii) The parts labelled **I, II** and **III** of device **A** when it is considered as a lever. [3 marks]
 I - Effort
 II - Load
 III - Pivot
- (iv) What the arrow represents in the device labelled **B** [1 mark]
Direction of effort / effort distance
- (v) The type of work done with **each** of the devices labelled:
 (α) **C**; Lifting objects
 (β) **D**; moving a vehicle or parts of an engine efficiently [2 marks]
- (d) (i) Names of the parts labelled **I, II, III, IV** and **V** [5 marks]
 I - Stomach
 II - Small intestines
 III - Large intestines
 IV - Rectum
 V - Oesophagus / gullet
- (ii) The part(s) of the digestive system where
 (α) digestion of food substances occur
 I and II
 (β) digested food is absorbed into the bloodstream [3 marks]
 II
- (iii) The end-products of the digestion that is absorbed into the bloodstream [3 marks]
 Amino acids, glucose, fatty acids and glycerol

2. (a) **Four** weather measuring instruments. [4 marks]
Barometer, rain gauge, anemometer, wind vane, hygrometer, sun dial / lightmeter
- (b) The stages in the life cycle of a mosquito. [4 marks]
Egg stage, larva stage, pupa stage and adult stage
- (c) (i) **Two** properties of water
- Colourless
 - Odourless
 - Tasteless
 - Boils at 100°C
 - Freezes at 0°C
- (ii) Why it is advisable to wash clothes with soft water [4 marks]
It lathers better with soap, since it does not contain dissolved salts such as calcium carbonate, magnesium hydroxide and calcium sulphate.
There is no production of scum, therefore it makes washing easier and faster.
- (d) **Three** ways in which soil profile is important. [3 marks]
- to determine the type of crop to grow
 - to determine the most suitable farming system to use
 - to determine the type / amount of fertilizer needed
 - to determine the cultural practices to use
 - to determine the type of tools / equipment to use
 - to determine the cost of production
3. (a) **Three** modes of heat transfer. [3 marks]
Conduction, convection and radiation
- (b) (i) What a *deficiency disease* is
A disease that results from the lack or shortage of certain nutrients in the body of an organism
- (ii) **Three** deficiency diseases in humans. [5 marks]
Kwashiorkor, goiter, rickets, anaemia, scurvy, night blindness
- (c) **Two** ways in which **each** of the following factors cause depletion of soil resources:
- (i) burning
- kills soil micro organism, such as nitrogen-fixing bacteria
 - makes the land bare, which results in soil erosion
 - causes faster evaporation of soil water

- destroys soil nutrients
- (ii) leaching [4 marks]
 - washes away water-soluble plant nutrients, especially nitrates and sulphur.
 - leads to soil acidity
 - affects the texture of the topsoil.
- (d) **Three** processes that can change matter from one state to another. [3 marks]
 - Freezing
 - Condensation
 - Sublimation
 - Melting
 - Evaporation
 - Deposition

4. (a) (i) What a *satellite* is.
A body that moves around / orbits a planet
- (ii) **Three** uses of artificial satellites. [5 marks]
 - weather forecasting
 - communication
 - scientific exploration
 - Navigation – using GPS, etc.
 - television/ radio broadcast
 - military purposes
- (b) The composition of **each** of the following alloys:
- (i) brass
copper and zinc
- (ii) steel
iron and carbon
- (iii) bronze [3 marks]
copper and tin
- (c) **Three** cultural practices used in vegetable production. [3 marks]
 - Weeding
 - Pruning
 - Mulching
 - Watering
 - Thinning out
 - Staking
- (d) **Four** parts of the respiratory system in humans. [4 marks]

Nostril, bronchus, bronchioles, alveoli, trachea, pharynx, blood capillaries

5. (a) (i) What a *force* is
A push or pull on an object
- (ii) **Two** ways in which forces could affect a body [4 marks]
A force can:
- cause a stationery body to move
 - cause a moving body to stop
 - increase the speed of a moving body (acceleration)
 - decrease the speed of a moving body (deceleration)
 - change the shape of a body

- (b) (i) What a *chemical change* is
A change in which a new substance is formed and is irreversible, eg, the neutralization reaction between HCl and NaOH

- (ii) **Three** differences between *chemical change* and *physical change*.

CHEMICAL CHANGE	PHYSICAL CHANGE
A new substance is formed	No new substance is formed
It is irreversible (not reversible)	It is usually reversible
Heat energy is given off or absorbed	No heat energy given off or absorbed
Chemical bonds are broken and new ones form	Chemical bonds are not broken and no new ones are formed

[5 marks]

- (c) **Three** physical properties of soil. [3 marks]

- Soil Texture,
- Soil Structure,
- Water-holding capacity/ permeability / porosity
- Soil colour
- Bulk density
- Soil temperature
- Soil Consistency / Soil strength

- (d) **Three** diseases of the circulatory system in humans. [3 marks]

- Anaemia
- Haemophilia
- Coronary artery disease
- Arteriosclerosis
- Leukemia
- stroke
- hypertension
- heart failure

- heart attack
- Aortic dissection
- Cardiomyopathy, etc

6. (a) The systematic name for **each** of the following compounds:

- (i) H_2O - Dihydrogen monoxide or hydrogen oxide
- (ii) MgO - Magnesium oxide
- (iii) CaO - Calcium oxide
- (iv) CaCl_2 - Calcium chloride

[4 marks]

(b) The instrument used in measuring the following:

- (i) length of a rope -
Metre rule or tape measure
- (ii) mass of a stone -
Beam balance, triple beam balance,
- (iii) temperature of a liquid-
Mercury-in-glass thermometer, alcohol thermometer
- (iv) volume of a liquid -
Measuring/graduated cylinder, beaker, pipette, volumetric flask, measuring cup/jug

[4 marks]

(c) **Three** factors that influence vegetable crop production.

[3 marks]

- Site selection
- Quality of crop variety /planting material
- Nature of soil
- Climate / weather conditions
- Cultural practices
- Harvesting
- Storage
- Marketing

(d) **Four** stages in the life cycle of a flowering plant.

[4 marks]

- Germination
- Seedling
- Mature plant
- Flowering
- Pollination
- Fertilization
- Formation of seeds and fruits
- Dispersal of seeds and fruits

JUNE 2015

INTEGRATED SCIENCE 1

OBJECTIVE TEST

45 minutes

1. Brass is an alloy of
 - A. iron and carbon
 - B. zinc and copper
 - C. iron and copper
 - D. copper and tin
2. An example of inorganic fertilizer is
 - A. ammonium nitrate
 - B. cow dung
 - C. farmyard manure
 - D. poultry dropping
3. The component of the human blood which transports oxygen to all parts of the body is
 - A. plasma
 - B. platelets
 - C. red blood cells
 - D. white blood cells
4. The importance of the fuse in an electrical circuit is to
 - A. regulate the voltage
 - B. prevent damage to electrical appliances
 - C. alter the flow of current in the circuit
 - D. minimize the use of current
5. Which of the following step(s) is/are required in the scientific method?
 - I. Formulation of hypothesis
 - II. Identification of the problem
 - III. Experimentation
 - A. I only
 - B. II only
 - C. II and III only
 - D. I, II and III
6. Which of the following food items produces amino acids as end product of digestion?
 - A. Cabbage
 - B. Fish
 - C. Margarine

- D. Rice
7. The process of increasing the strength of a signal using a transistor is known as
- A. amplification
 - B. biasing
 - C. doping
 - D. switching
8. The physical arrangement of soil particles into aggregates is termed
- A. soil porosity
 - B. soil profile
 - C. soil structure
 - D. soil texture
9. The solvent which is **most** effective in washing bitumen from the hand is
- A. acid
 - B. alcohol
 - C. kerosene
 - D. water
10. A piece of stone could be classified as an opaque material because it
- A. absorbs all the light incident on it
 - B. does not absorb light incident on it
 - C. allows all the light incident on it to pass through it
 - D. does not allow light incident on it to pass through it
11. Fish swims in water with little resistance because it possesses
- A. gills
 - B. scales
 - C. caudal fins
 - D. streamlined body
12. Which of the following conditions promote(s) rusting of iron?
- I. Air
 - II. Moisture
 - III. Oil
- A. I only
 - B. II only
 - C. I and II only
 - D. II and III only
13. The change in the volume of water when a piece of stone is dropped into it is equal to the
- A. density of the stone
 - B. mass of the stone
 - C. volume of the stone
 - D. weight of the stone

14. The second stage in the life cycle of a mosquito is the
- A. egg
 - B. imago
 - C. larva
 - D. pupa
15. Which of the following objects could be attracted by a magnet?
- A. Copper chain
 - B. Gold chain
 - C. Steel blade
 - D. Aluminium spoon
16. The knowledge of soil texture is important because it
- A. influences plant population
 - B. determines the planting distance
 - C. determines the type of crop to be grown
 - D. influences the method of pest control
17. How many atoms are present in CaCl_2 ?
- A. 2
 - B. 3
 - C. 4
 - D. 5
18. The transfer of heat from the bottom to the top of a beaker containing water is by
- A. absorption
 - B. conduction
 - C. convection
 - D. radiation
19. An example of a chemical compound is
- A. aluminium
 - B. ammonia
 - C. oxygen
 - D. phosphorus
20. The type of the human teeth used for biting food substances is
- A. canines
 - B. incisors
 - C. molars
 - D. premolars
21. Which of the following heavenly bodies is a star?
- A. Jupiter
 - B. Moon
 - C. Sun

- D. Venus
22. Which of the following organisms is an endoparasite?
- A. Fleas
 - B. Louse
 - C. Tapeworm
 - D. Tick
23. The part of the plant where pollination occurs is the
- A. flowers
 - B. leaves
 - C. roots
 - D. stems
24. Which of the following substances is capable of neutralizing an acid?
- A. Vinegar
 - B. Water
 - C. Sodium chloride
 - D. Sodium hydroxide
25. Droplet infection is the method by which
- A. diseases are transmitted from animals to plants
 - B. infected animals are processed
 - C. human beings protect themselves from diseases
 - D. diseases are spread through the air
26. An example of a derived quantity is
- A. length
 - B. mass
 - C. time
 - D. velocity
27. A structure in the human reproductive system which stores sperms temporarily is the
- A. epididymis
 - B. penis
 - C. scrotum
 - D. vulva
28. Which of the following statements about diffusion is/are **correct**?
- I. It involves the movement of water molecules only
 - II. It occurs in both gases and liquids
 - III. It involves a semi-permeable membrane
- A. I only
 - B. II only
 - C. I and II only
 - D. I, II and III

29. A farming system which requires **at least** three years programme is
- A. mixed cropping
 - B. crop rotation
 - C. organic farming
 - D. mixed farming
30. The component of a living cell responsible for respiration is
- A. chloroplast
 - B. mitochondria
 - C. nucleus
 - D. vacuole
31. Which of the following statements about a second class lever is correct? The
- A. pivot is between the load and the effort
 - B. pivot and the load are at the same position
 - C. load is between the effort and the pivot
 - D. effort is between the load and the pivot
32. One advantage of soft water over hard water is that soft water
- A. has a pleasant taste
 - B. can prevent heart diseases
 - C. does not waste soap
 - D. forms scales in kettles
33. The gas produced when glucose is oxidized during aerobic respiration is
- A. hydrogen
 - B. nitrogen
 - C. oxygen
 - D. carbon dioxide
34. The systematic name for N_2O is
- A. nitrogen (I) oxide
 - B. nitrogen (II) oxide
 - C. nitrogen (III) oxide
 - D. nitrogen (IV) oxide
35. A husbandry practice which involves the removal of unwanted plant parts is known as
- A. mulching
 - B. pruning
 - C. thinning
 - D. pricking out
36. The number of p-n junctions in a transistor is
- A. 1
 - B. 2
 - C. 3

D. 4

37. Fruits dispersed by wind

- A. have hairs
- B. are sticky
- C. are succulent
- D. have hooks

38. Substances that burn living tissues when they come into contact with these tissues are considered

- A. corrosive
- B. flammable
- C. irritant
- D. toxic

39. The function of the platelets in the circulatory system of humans is to

- A. transport oxygen
- B. transport carbon dioxide
- C. clot the blood
- D. defend the body

40. Energy is the

- A. force to move an object
- B. weight to do work
- C. ability to do work
- D. power to carry an object

JUNE 2015

INTEGRATED SCIENCE 1

OBJECTIVE TEST

SOLUTIONS

1. B. zinc and copper
2. A. ammonium nitrate
3. C. red blood cells
4. B. prevent damage to electrical appliances
5. D. I, II and III
6. B. Fish
7. A. amplification
8. B. soil structure
9. C. kerosene
10. D. does not allow light incident on it to pass through it
11. D. streamlined body
12. C. I and II only
13. C. volume of the stone
14. C. larva
15. C. Steel blade
16. C. determines the type of crop to be grown
17. B. 3
18. C. convection
19. B. ammonia
20. B. incisors
21. C. Sun
22. C. Tapeworm
23. A. flowers
24. D. Sodium hydroxide
25. D. diseases are spread through the air
26. D. velocity
27. A. epididymis
28. B. II only
29. B. crop rotation

- 30. B. mitochondria
- 31. C. load is between the effort and the pivot
- 32. C. does not waste soap
- 33. D. carbon dioxide
- 34. D. nitrogen (I) oxide
- 35. B. pruning
- 36. B. 2
- 37. A. have hairs
- 38. A. corrosive
- 39. C. clot the blood
- 40. C. ability to do work

JUNE 2015 (FIRST SITTING)

INTEGRATED SCIENCE 2

ESSAY

1 ¼ hours

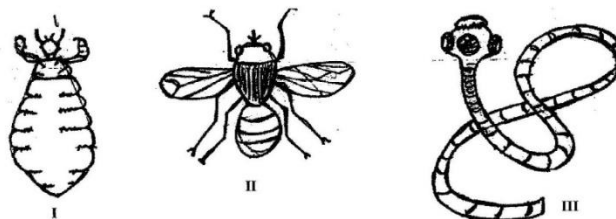
This paper is in two parts: I and II. Answer Question 1 in part I and any other four questions in part II

PART I **[40 marks]**

Answer **all** of Question 1

1. (a) The diagrams below are illustrations of three different organisms harmful to farm animals

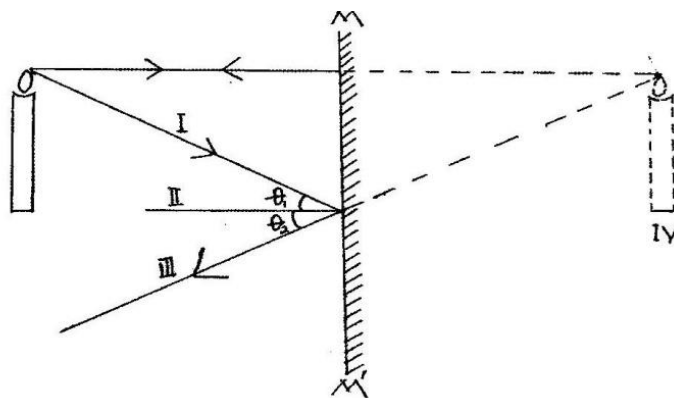
Study the diagrams carefully and answer the questions that follow



- (i) Identify **each** of the organisms labelled **I**, **II** and **III** [3 marks]
- (ii) Which of the organisms is/are:
- (α) parasite(s)
 - (β) pest(s) [3 marks]
- (iii) State **one** effect **each** of the following organism on farm animals
- (α) **I**;
 - (β) **II**;
 - (γ) **III**. [3 marks]
- (iv) State **three** methods of controlling the organism labelled **III** [3 marks]

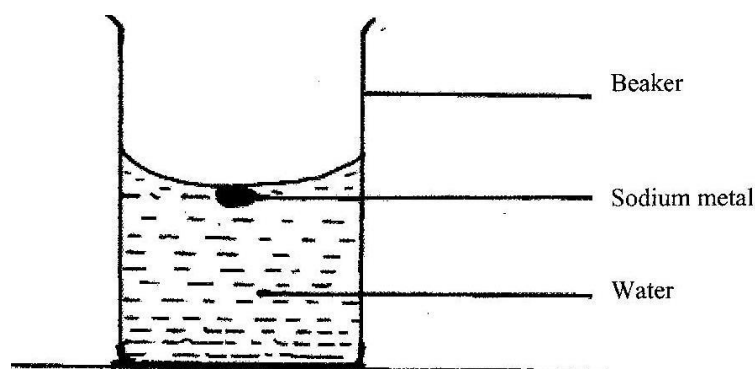
- (b) The diagram below illustrates a burning candle placed in front of a plane mirror MM' and an image of the candle formed in the mirror.

Study the diagram carefully and answer the questions that follow



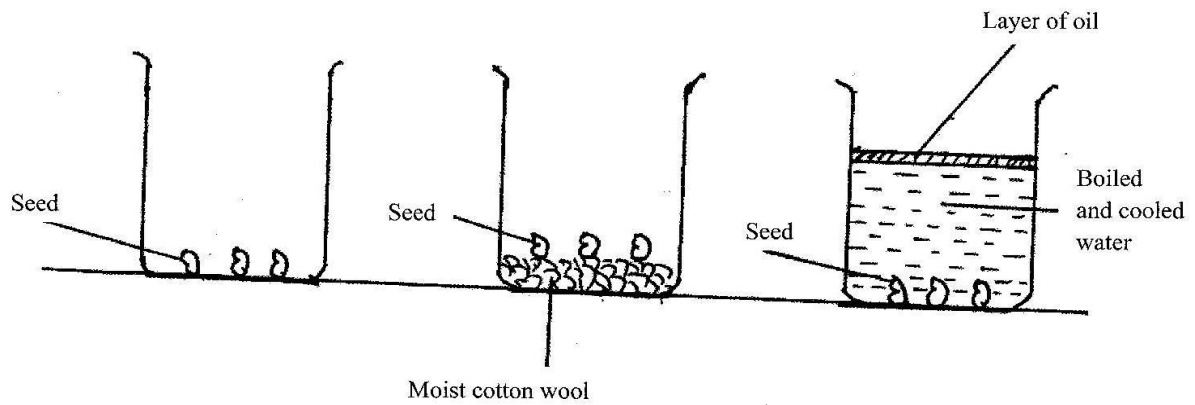
- (i) Name the parts labelled **I**, **II**, **III** and **IV** [4 marks]
 - (ii) State the relationship between angles θ_1 and θ_2 [1 mark]
 - (iii) Give **three** characteristics of **IV** in the diagram [3 marks]
 - (iv) Explain why **IV** is represented in broken lines [2 marks]
- (c) In an experiment to investigate the reactivity of sodium, a piece of sodium metal was dropped in a beaker containing water. The experimental set-up is as illustrated below.

Study the set-up carefully and answer the questions that follow



- (i) State what would happen if a glowing splint was held at the mouth of the beaker [2 marks]
 - (ii) Name the gas evolved [1 mark]
 - (iii) Write a balanced chemical equation for the reaction that occurred [3 marks]
 - (iv) Name **two** other metals that can react in a similar ways as the sodium [2 marks]
- (d) An experiment to investigate the conditions for germination of viable bean seeds was carried out. The diagrams below are illustrations of the different conditions in which the seeds were kept. The beakers labelled **A**, **B** and **C** containing the seeds were kept at room temperature.

Study the diagrams carefully and answer the questions that follow

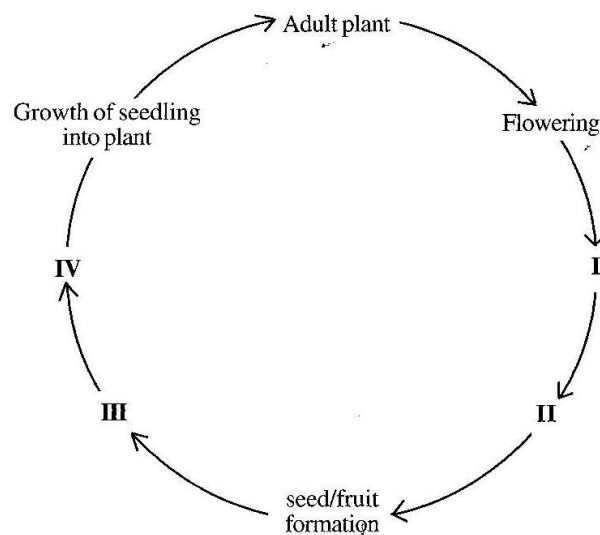


- (i) State what would happen to the seeds in **each** of the beakers labelled **A, B** and **C** when the experiment was observed after five days. [3 marks]
- (ii) Give reasons for **each** of your answers in (i) [4 marks]
- (iii) Why was oil spread on the surface of the water in the beaker labelled **C**? [2 marks]

PART II [60 marks]

Answer **four** questions **only** from this section

2. (a) (i) What is *weather*? [4 marks]
- (ii) State **two** differences between *weather* and *season* [4 marks]
- (b) State the composition of **each** of the following alloys; [3 marks]
- (i) steel;
- (ii) stainless steel
- (c) List **four** benefits of vegetables to humans [4 marks]
- (d) The diagram below is an illustration of life-cycle of a flowering plant.



Name **each** of the stages labelled **I, II, III** and **IV**

[4 marks]

3. (a) Explain how
 (i) lithium atom becomes **positively** charged
 (ii) oxygen atom becomes **negatively** charged [2 marks]
- (b) (i) What is *potential energy*?
 (ii) A coconut of mass 2 kg is on a tree 5 m tall. Determine the potential energy of the coconut at this height [Take $g = 10 \text{ ms}^{-2}$] [5 marks]
- (c) State **four** causes of teenage pregnancy [4 marks]
- (c) State **four** uses of soil in agriculture. [4 marks]
4. (a) Give **four** health benefits of water to humans [4 marks]
- (b) (i) State **two** ways in which crop rotation is important in crop production
 (ii) Distinguish between *mixed cropping* and *mixed farming* [4 marks]
- (c) (i) What is a *fuse*?
 (ii) Name the colour code of the wire on which a fuse is placed in a three-pin plug [3 marks]
- (d) (i) In a tabular form, state **three** differences between *osmosis* and *diffusion*
 (ii) State **one** way in which osmosis is similar to diffusion [4 marks]
5. (a) (i) What is a *balanced* ration in animal nutrition?
 (ii) State **two** benefits of balanced ration to poultry [4 marks]
- (b) List **four** hazards that may be encountered in teaching and learning of science [4 marks]
- (c) Name the parts of the circulatory system of humans [3 marks]
- (d) (i) What is a *simple machine*?
 (ii) State **two** methods of overcoming friction in everyday activities [4 marks]
6. (a) (i) What is a *magnetic field*?
 (ii) To which class of mixture does **each** of the following belong?
 (α) Smoke
 (β) Air
 (γ) Bronze [5 marks]
- (b) (i) What is *plaque* in human dentition?

- (ii) State the function of chlorophyll in photosynthesis [3 marks]
- (c) Identify the scientific principle underlying the operation of **each** of the following industries
- (i) kenkey production
 - (ii) salt smoking
 - (iii) fish smoking
 - (iv) biogas production [4 marks]
- (d) (i) What is *plant parasite*?
- (ii) Give **one** example of a plant parasite [3 marks]

JUNE 2015 (FIRST SITTING)

INTEGRATED SCIENCE 2

ESSAY

SOLUTIONS

PART I
[40 marks]

1. (a)
- (i) Identify **each** of the organisms labelled **I, II** and **III** [3 marks]
- I. - Louse
II. - Tsetse fly
III. - Tapeworm
- (ii) Which of the organisms is/are:
- (α) parasite(s)
Louse and tapeworm
- (β) pest(s) [3 marks]
tsetsefly
- (iii) State **one** effect **each** of the following organism on farm animals
- (α) **I:** Sucks blood of animals, resulting in anaemia
(β) **II:** Infects animals with sleeping sickness, which can lead to death
(γ) **III.** feeds on the digested food nutrients in the body of animal, which affects the immune system and leads to weakness and ill-health [3 marks]
- (iv) State **three** methods of controlling the organism labelled **III** [3 marks]
- Deworming,
 - regular checks by veterinary personnel
 - Proper disposal of sewage
 - clean grazing on low risk pasture
- (b)
- (i) Name the parts labelled **I, II, III** and **IV** [4 marks]
- I. - Incident ray
II. - Normal
III. - Reflected ray
IV. - Image
- (ii) State the relationship between angles θ_1 and θ_2 [1 mark]

Angles θ_1 and θ_2 are equal

(iii) Give **three** characteristics of **IV** in the diagram [3 marks]

- it is erect (upright)
- it is virtual
- It has the same size as the object
- Its distance to the mirror equals the distance of the object from the mirror
- It is formed behind the mirror

(iv) Explain why **IV** is represented in broken lines [2 marks]

IV is represented in broken lines because it is a virtual (not a real) image (cannot be captured on a screen)

(c) (i) State what would happen if a glowing splint was held at the mouth of the beaker [2 marks]

The glowing splint would be ignited (become aflame / catch fire)

(ii) Name the gas evolved [1 mark]

Hydrogen

(iii) Write a balanced chemical equation for the reaction that occurred [3 marks]



(iv) Name **two** other metals that can react in a similar ways as the sodium [2 marks]

Lithium, Potassium

(d)

(i) State what would happen to the seeds in **each** of the beakers labelled **A**, **B** and **C** when the experiment was observed after five days.

[3 marks]

Beaker A	-	Seeds do not germinate
Beaker B	-	Seeds germinate
Beaker C	-	Seeds do not germinate

(ii) Give reasons for **each** of your answers in (i) [4 marks]

- Seeds in beaker A do not germinate because there is no water / moisture present
- Seeds in beaker B germinate because all the conditions necessary for germination, namely, moisture, air, optimum temperature and viable seed are present.
- Seeds in beaker C do not germinate because there is no air present (due to layer of oil on water surface)

(iii) Why was oil spread on the surface of the water in the beaker labelled **C**? [2 marks]

To prevent air (atmospheric oxygen) from getting dissolved in the water and reaching the seeds

PART II
[60 marks]

Answer **four** questions **only** from this section

2. (a) (i) What is *weather*?

The atmospheric condition of a place at a particular time.

or

The state of the atmosphere at a particular place and time

or

The condition of the atmosphere of a place over a short period of time

- (ii) State **two** differences between *weather* and *season*

[4 marks]

WEATHER	SEASON
Atmospheric condition of a place over a short period of time	The average atmospheric condition of a place over a longer period of time within a year
Changes relatively quickly (lasts for a short time, usually about a day)	Changes relatively slowly (lasts for a longer time, usually 3 or more months)
It is less predictable	It is more predictable

- (b) State the composition of **each** of the following alloys;

- (i) steel;

iron and carbon

- (ii) stainless steel

[3 marks]

iron, carbon and chromium

- (c) List **four** benefits of vegetables to humans

[4 marks]

Provide mineral salts, which supports metabolic activities for proper functioning of the body

Provide vitamins for protection against diseases

Provide dietary fibre for easy bowel movement

Provide antioxidants, which fights stress and strengthen immune system

Makes our food tastier / more enjoyable

- (d) Name **each** of the stages labelled **I, II, III** and **IV**

[4 marks]

I. - Pollination

II. - Fertilization

III. - Dispersion / dispersal

IV. - Germination

3. (a) Explain how

(i) lithium atom becomes **positively** charged

Lithium has 1 electron in its outermost shell, making it unstable. In order to become stable, it loses the electron on its outermost shell. As a result, the atom becomes positively charged, since there is now 1 more positive charge (proton) than negative charge (electron)

(ii) oxygen atom becomes **negatively** charged

[2 marks]

Oxygen has 6 electrons in its outermost shell, making it unstable. In order to become stable, it gains 2 more electrons on its outermost shell to make it completely filled. As a result, the negative charges (electrons) become 2 more than the positive charges (protons), making the atom negatively charged.

(b) (i) What is *potential energy*?

The energy possessed by a body as a result of its position or state.

(ii) A coconut of mass 2 kg is on a tree 5 m tall. Determine the potential energy of the coconut at this height

[Take $g = 10 \text{ ms}^{-2}$]

[5 marks]

$$\begin{aligned}\text{Potential energy} &= \text{mass}(m) \times \text{height}(h) \times \text{acceleration due to gravity}(g) \\ &= 2 \text{ kg} \times 5 \text{ m} \times 10 \text{ ms}^{-2} \\ &= 100 \text{ kgms}^{-2} \\ &= 100 \text{ J}\end{aligned}$$

(c) State **four** causes of teenage pregnancy

[4 marks]

- (i) Indiscriminate / Casual sex
- (ii) Irresponsible parenting
- (iii) Lack of sex education in schools
- (iv) Illiteracy
- (v) Negative peer pressure
- (vi) Lack of contraceptive use / rejection of family planning methods
- (vii) Wrong application of birth control measures
- (viii) Unemployment
- (ix) Poverty
- (x) Streetism

(d) State **four** uses of soil in agriculture.

[4 marks]

- It serves as a support to agricultural crops
- It serves as a habitat (home) for several micro-organisms, which are important for plant growth.
- It holds water for the use of agricultural crops and animals.
- It contains mineral salts which are used by crops for healthy growth.
- It supports grass and other plants, which farm animals feed on for their sustenance
- It helps maintain suitable temperature for the development of plant roots and micro-organisms.

4. (a) Give **four** health benefits of water to humans [4 marks]

- It helps with easy bowel movement (Prevents constipation)
- It helps with the metabolic activities of the body
- It can relieve stress
- It can relieve headaches
- It promotes healthy and radiant skin
- It helps to maintain optimum blood pressure
- It helps to maintain suitable body temperature.
- It supports the proper functioning of the cells/ tissues/ organs of the body.

- (b) (i) State **two** ways in which crop rotation is important in crop production

- It helps in maintaining soil fertility
- The legumes included in crop rotation fix atmospheric nitrogen in the soil,
- The growing of cover crops helps check soil erosion
- The system of rotation helps check pests and diseases
- There is a reduced risk of losing yield, in case of pest or disease attack.

- (ii) Distinguish between *mixed cropping* and *mixed farming* [4 marks]

Mixed cropping – Different types of crops are grown on the same piece of land

Mixed farming – Crops are grown and farm animals are reared at different sections of the same piece of land.

- (c) (i) What is a *fuse*?

An electrical safety device containing a piece of a metal that melts if the current running through it exceeds a particular level, thereby breaking the circuit

or

a safety device consisting of a strip of wire that melts and breaks an electric circuit if the current exceeds a safe level

- (ii) Name the colour code of the wire on which a fuse is placed in a three-pin plug

Red/ brown

[3 marks]

- (d) (i) In a tabular form, state **three** differences between *osmosis* and *diffusion*

OSMOSIS	DIFFUSION
It involves the movement of only solvent or water molecules	It involves movement of solute molecules /particles
Semi-permeable membrane present	No semi-permeable membrane
Applies in only liquids (solvent molecules)	Applies in solids, liquids and gases
Movement of solvent molecules from lower to higher	Movement of solute particles from a higher concentration to a

concentration	lower concentration
---------------	---------------------

- (ii) State **one** way in which osmosis is similar to diffusion [4 marks]
- They both involve the movement of molecules
 - The particles / molecules move to cause equality in the concentrations of the regions
 - presence of a concentration gradient
5. (a) (i) What is a *balanced* ration in animal nutrition?
The feed allowed for a given animal during a 24-hour period which contains all the essential nutrients to the animal in the right proportion and amount .
- (ii) State **two** benefits of balanced ration to poultry [4 marks]
- It ensures healthy and strong birds
 - It increase the quality of the eggs produced
 - It increases the quantity of eggs produced
 - It increases the rate of growth
- (b) List **four** hazards that may be encountered in teaching and learning of science [4 marks]
- Corrosion of one's skin by corrosive chemicals
 - Burns from open flame or explosives
 - Injuries sustained from falling as a result of slippery floor
 - Health risk from inhaling of poisonous gases
 - Health risk from ingestion of toxic chemicals
 - Skin irritation from contact with irritants
 - Health risks from Electrical shocks
 - Eye irritation from strong light or radiation or irritants
- (c) Name the parts of the circulatory system of humans [3 marks]
- Heart,
 - blood
 - Blood vessels (arteries, veins, capillaries)
- (d) (i) What is a *simple machine*?
A mechanical device that makes work easier and/ or faster
- (ii) State **two** methods of overcoming friction in everyday activities [4 marks]
- Applying lubricants such as grease / oil
 - Smoothing the contact surfaces
 - Using wheels / rollers
 - Using ball bearings
6. (a) (i) What is a *magnetic field*?

A region of space surrounding a magnet or current-carrying circuit in which the resulting magnetic force can be detected

(ii) To which class of mixture does **each** of the following belong?

- | | | | |
|-----|--------|---|---------------------|
| (α) | Smoke | - | solid-gas mixture |
| (β) | Air | - | gas-gas mixture |
| (γ) | Bronze | - | solid-solid mixture |

[5 marks]

(b) (i) What is *plaque* in human dentition?

A sticky, colorless deposit of bacteria that is constantly forming on the tooth surface

OR

A sticky, colorless film of bacteria that constantly forms on our teeth and along the gum line

(ii) State the function of chlorophyll in photosynthesis

[3 marks]

It traps/absorbs sunlight, which provides energy for the manufacture of food

(c) Identify the scientific principle underlying the operation of **each** of the following industries

- | | | | |
|-------|-------------------|---|-------------------|
| (i) | kenkey production | - | fermentation |
| (ii) | salt making | - | evaporation |
| (iii) | fish smoking | - | food preservation |
| (iv) | biogas production | - | fermentation |

[4 marks]

(d) (i) What is *plant parasite*?

A plant that derives some or all of nutrients and water from another living organism

(ii) Give **one** example of a plant parasite

[3 marks]

dodder, mistletoe, the corpse flower, cactus mistletoe

JUNE 2015 (SECOND SITTING)

INTEGRATED SCIENCE 2

ESSAY

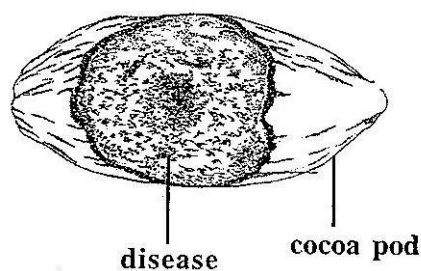
1 ¼ hours

This paper is in two parts: I and II. Answer Question 1 in part I and any other four questions in part II

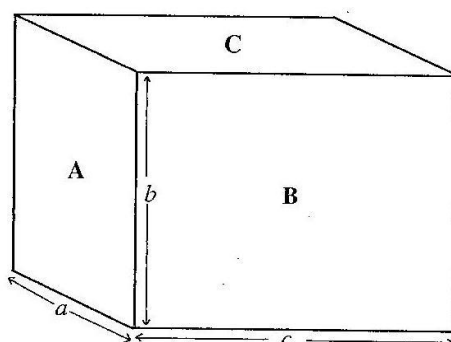
PART I **[40 marks]**

Answer **all** of Question 1

1. (a) The diagram below is an illustration of a cocoa pod infected by a disease.
Study the diagram carefully and answer the questions that follow



- (i) Name the disease illustrated on the pod [1 mark]
(ii) Mention the causative organism of the disease [1 mark]
(iii) Describe **one** symptom of the disease [2 mark]
(iv) State **one** farm practice that promotes the spread of the disease [1 mark]
(v) Give **two** practices farmers could adopt to control the disease [2 mark]
- (b) The diagram below is an illustration of a cuboid with faces **A**, **B** and **C** shown and sides *a*, *b* and *c* indicated.
Study the diagram carefully and answer the questions that follow



(i) Measure and record the length of **each** of the sides labelled **a**, **b** and **c**, in centimetres.

[3 marks]

(ii) Determine the area of **each** of the faces **A** and **B**

[4 marks]

(iii) If the weight of the cuboid is 100 N, determine the pressure exerted by the cuboid in N/cm² when it lies on

(α) face **A**

(β) face **B**.

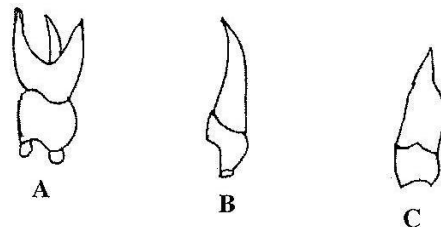
[4 marks]

(iv) From your results in (iii), explain how pressure exerted by a body changes with the area on which it lies

[2 marks]

(c) The diagrams below are illustrations of three different types of teeth, labelled **A**, **B** and **C**, in humans

Study the diagram carefully and answer the questions that follow



(i) Name **each** type of teeth labelled **A**, **B** and **C**.

[3 marks]

(ii) State **one** function of **each** type of teeth labelled **A**, **B** and **C**.

[3 marks]

(iii) State **two** observable differences between the tooth labelled **A** and that labelled **B**

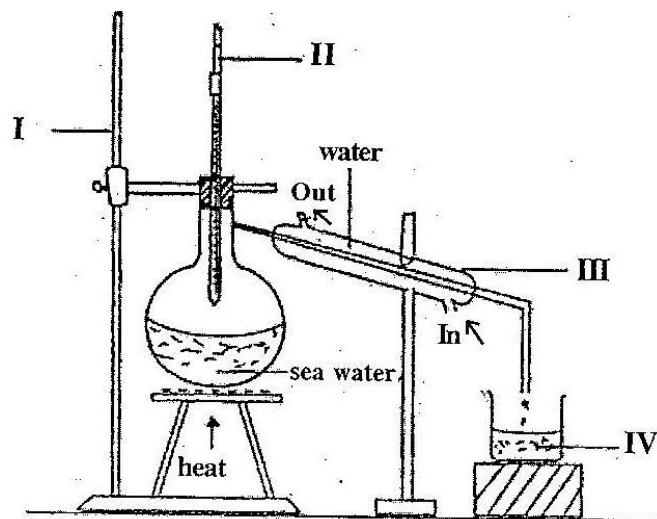
[2 marks]

(iv) Mention **two** ways of ensuring proper dental care.

[2 marks]

(d) The diagram below is an illustration of an experimental set-up used to separate water and salt from salt solution.

Study the diagram carefully and answer the questions that follow



- (i) Name **each** of the parts labelled **I, II, III** and **IV**. [4 marks]
- (ii) Name the method of separation used. [1 marks]
- (iii) State **one** function of **each** of the parts labelled **II** and **III** [2 marks]
- (iv) Name **three** physical processes that are involved in the separation of the mixture. [3 marks]

PART II

[60 marks]

Answer **four** questions **only** from this section

2. (a) (i) What is *technology*? [4 marks]
 (ii) State **two** differences between *science* and *technology*. [4 marks]
- (b) Draw potassium atom and show the distribution of electrons in its shells. [K = 19] [4 marks]
- (c) What energy transformations take place in **each** of the following activities?
 (i) Using a flashlight battery to produce light in a bulb.
 (ii) Using a microphone to address a gathering. [4 marks]
- (d) List **three** components of soil. [3 marks]
3. (a) (i) Mention the junctions of a transistor.
 (ii) Draw the circuit symbol for *n-p-n* transistor [4 marks]
- (b) State **one** function of **each** of the following parts of the human circulatory system:
 (i) blood
 (ii) blood vessel
 (iii) heart [3 marks]

(c) Classify **each** of the following substances as element, compound or mixture:

- (i) air
- (ii) brass
- (iii) chlorine
- (iv) gold
- (v) water

[5 marks]

(d) State **three** ways of caring for farm machines

[3 marks]

4. (a) (i) State **two** physical properties of water

(ii) Using litmus paper, explain why water is neutral.

[3 marks]

(b) Give **one** function of **each** of the following nutrients in plants

- (i) nitrogen
- (ii) potassium

[2 marks]

(c) (i) State **two** human activities that disrupt the carbon cycle.

(ii) State **two** effects of the disruption of the carbon cycle on the environment. [4 marks]

(d) (i) What is *density* of a body?

(ii) A body of mass 50 kg has a density of 2 kg/m^3 . Calculate the volume of the body.

[5 marks]

5. (a) What is a *vegetable crop*?

[2 marks]

(b) (i) What is air *pollution*?

(ii) State **two** human activities that lead to air pollution.

[4 marks]

(c) (i) List **three** modes of heat transfer

(ii) Name the type of medium in which **each** mode of heat transfer you have listed in (i) takes place

[6 marks]

(d) Give **three** reasons why proteins are important in the diet of mammals.

[3 marks]

6. (a) State **one** function of **each** of the following parts of the human reproductive system

- (i) penis
- (ii) ovary
- (iii) oviduct
- (iv) testis
- (v) uterus

[5 marks]

(b) (i) What is *electrical conductor*?

- (ii) List **two** substances which are insulators. [4 marks]
- (c) State **three** ways of maintaining soil fertility. [3 marks]
- (d) Write a word equation for **each** of the following chemical reactions:
- (i) oxygen and hydrogen
 - (ii) nitrogen and hydrogen
 - (iii) sodium and chlorine [3 marks]

END OF PAPER

JUNE 2015 (SECOND SITTING)

INTEGRATED SCIENCE 2

ESSAY

SOLUTIONS

PART I
[40 marks]

1. (a)

- (i) Name the disease illustrated on the pod [1 mark]
Black pod disease
- (ii) Mention the causative organism of the disease [1 mark]
Fungus (*Phytophthora* spp. pathogens)
- (iii) Describe **one** symptom of the disease [2 mark]
Little yellow spots, which turn brown and enlarge to a dark brown or black lesion within a few days
- (iv) State **one** farm practice that promotes the spread of the disease [1 mark]
- Late harvesting of ripe pods
 - Insufficient spacing of tree
 - Allowing infected pods to remain on trees for a long time
- (v) Give **two** practices farmers could adopt to control the disease [2 mark]
- Cultural control practices**
- Removal of infected pods
 - Wider spacing of trees
 - Pruning of trees
 - More frequent ripe pod harvesting
- Chemical control practice**
Application of fungicide

(b)

- (i) Measure and record the length of **each** of the sides labelled **a**, **b** and **c**, in centimetres. [3 marks]
- Length of side a = 2 cm
Length of side b = 4 cm
Length of side c = 5 cm

- (ii) Determine the area of **each** of the faces **A** and **B** [4 marks]

$$\begin{aligned}\text{Area of face A} &= \text{Side a} \times \text{Side b} \\ &= 2 \text{ cm} \times 4 \text{ cm} \\ &= 8 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Area of face B} &= \text{Side b} \times \text{Side c} \\ &= 4 \text{ cm} \times 5 \text{ cm} \\ &= 20 \text{ cm}^2\end{aligned}$$

- (iii) If the weight of the cuboid is 100 N, determine the pressure exerted by the cuboid in N/cm² when it lies on

(α) face **A**

$$\begin{aligned}\text{Pressure on A} &= \frac{\text{Force}}{\text{Area}} = \frac{100 \text{ N}}{8 \text{ cm}^2} \\ &= \frac{25 \text{ N}}{2 \text{ cm}^2} = \underline{12.5 \text{ N/cm}^2}\end{aligned}$$

(β) face **B**.

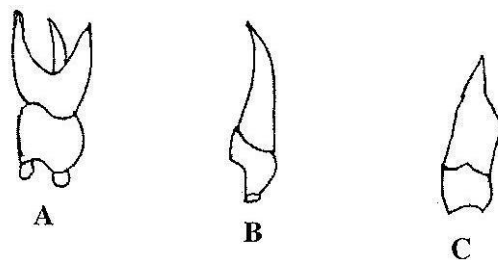
$$\text{Pressure on B} = \frac{100 \text{ N}}{20 \text{ cm}^2} = \underline{5 \text{ N/cm}^2} \quad [4 \text{ marks}]$$

- (iv) From your results in (iii), explain how pressure exerted by a body changes with the area on which it lies [2 marks]

- From the results in (iii) above, it is clear that pressure increases with decreasing area over which it is exerted, given that the force applied is constant.
- In other words, given a constant force, the greater the surface area over which the force is applied, the lesser the pressure; and the lesser the surface area, the greater the pressure.
- In other words, where force is constant, pressure is inversely proportional to the surface area over which it is applied.

- (c) The diagrams below are illustrations of three different types of teeth, labelled **A**, **B** and **C**, in humans

Study the diagram carefully and answer the questions that follow



- (i) Name **each** type of teeth labelled **A**, **B** and **C**. [3 marks]
A - molar

- B - canine
C - premolar

(ii) State **one** function of **each** type of teeth labelled **A**, **B** and **C**. [3 marks]

- A - crushing or chewing or grinding food
B - holding/gripping or tearing food
C - holding/gripping or crushing food

(iii) State **two** observable differences between the tooth labelled **A** and that labelled **B** [2 marks]

<u>Tooth A</u>	<u>Tooth B</u>
3 pointed projections (cusps)	Only 1 pointed projection (cusp)
Broad or wide	Narrow

(iv) Mention **two** ways of ensuring proper dental care. [2 marks]

- Brushing teeth after meals at least twice a day
- Avoid taking very hot or very cold foods or drinks
- Avoid taking lots of sweets, especially ones that stick to the teeth
- Avoid picking teeth with very sharp/ hard objects
- Regularly eat foods rich in calcium, phosphorus and vitamins
- Visit the dentist regularly for check ups

(d)

(i) Name **each** of the parts labelled **I**, **II**, **III** and **IV**. [4 marks]

- I.** - Retort stand
II. - Thermometer
III. - Condenser
IV. - pure water (distillate)

(ii) Name the method of separation used. [1 marks]
Distillation

(iii) State **one** function of **each** of the parts labelled **II** and **III** [2 marks]

- II** - Measures the temperature within the flask
III - It cools the steam (water vapour) or condenses the steam

(iv) Name **three** physical processes that are involved in the separation of the mixture. [3 marks]
Evaporation/ vapourisation, condensation, crystallisation

PART II

[60 marks]

*Answer **four** questions **only** from this section*

2. (a) (i) What is *technology*?

The use of scientific knowledge to solve problems in everyday life

Or

The application of scientific knowledge and methods to make life / work easier, faster and more comfortable

Or

The study, development, and application of devices, machines, and techniques for manufacturing and productive processes

(ii) State **two** differences between *science* and *technology*.

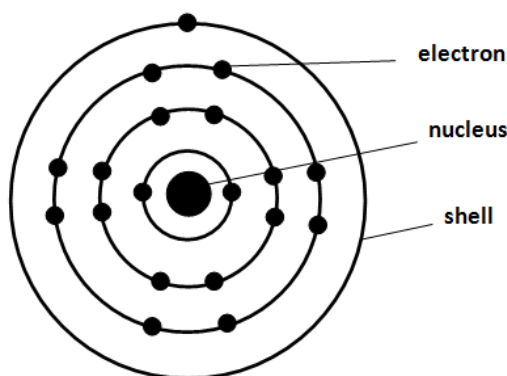
[4 marks]

SCIENCE	TECHNOLOGY
Aims at gaining knowledge about nature	Aims at applying scientific knowledge to solve problems
Focuses more on experimentation and analysis	Focuses more on synthesis of design
Mainly theory based	Mainly practical based
Generally cannot be used to solve everyday problems	Are generally used to solve everyday problems

(b) Draw potassium atom and show the distribution of electrons in its shells.

[K = 19]

[4 marks]



(c) What energy transformations take place in **each** of the following activities?

(i) Using a flashlight battery to produce light in a bulb.

Chemical energy → Electrical energy → Light energy and heat

(ii) Using a microphone to address a gathering.

[4 marks]

Electrical energy → Sound energy

(d) List **three** components of soil.

[3 marks]

- Mineral matter / rock particles,
- organic matter/humus,
- water,

- air,
- micro-organisms

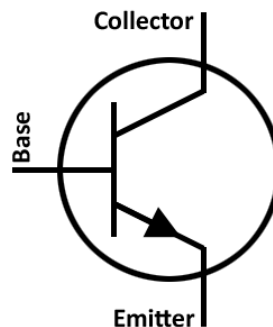
3. (a) (i) Mention the junctions of a transistor.

p-n junction

n-p junction

(ii) Draw the circuit symbol for *n-p-n* transistor

[4 marks]



(b) State **one** function of **each** of the following parts of the human circulatory system:

(i) blood

- Transports oxygen to the cells of the body
- Transports metabolic waste materials away from the cells of the body
- Transports digested food nutrients to parts of the body
- Helps to maintain a uniform body temperature
- Prevents excessive blood loss by clotting.
- Helps to protect the body against infections and diseases

(ii) blood vessel

- Carries blood from the heart to/from the parts of the body
- Helps in controlling the blood pressure within the body

(iii) heart

Pumps blood to the parts of the body

[3 marks]

(c) Classify **each** of the following substances as element, compound or mixture:

- | | | |
|----------------|---|----------|
| (i) air | - | mixture |
| (ii) brass | - | mixture |
| (iii) chlorine | - | element |
| (iv) gold - | | element |
| (v) water | - | compound |

[5 marks]

(d) State **three** ways of caring for farm machines

[3 marks]

- Oiling or greasing metal parts
- Washing and drying after use
- Tightening bolts and nuts when loose
- Sharpening cutting edges when blunt
- Using them for their intended purposes only

4. (a) (i) State **two** physical properties of water
- Colourless,
 - odourless
 - tasteless
 - freezes at 0° Celsius
 - boils at 100° Celsius
 - density of 1 g/cm³ (at 4°C)
 - Among liquids, is a good conductor of heat
 - Universal solvent
 - High surface tension
- (ii) Using litmus paper, explain why water is neutral. [3 marks]
- Blue litmus paper turns red in acidic medium and red litmus paper turns blue in basic medium. The colour of litmus paper, whether red or blue, remains the same in the presence of water. This shows that water is neutral, meaning it is neither acidic nor basic. (Pure water has a pH of 7)
- (b) Give **one** function of **each** of the following nutrients in plants
- (i) nitrogen
- Essential for photosynthesis (carbohydrate production), due to its presence in chlorophyll
 - Promotes rapid growth of plants, as a component of plant protein.
- (ii) potassium [2 marks]
- Regulates the opening & closing of stomata
 - Enables the plants to withstand extreme temperatures
 - It increases the plants' ability to use water more efficiently
 - Facilitates the production of starch from sugars
 - Essential for the synthesis of protein
- (c) (i) State **two** human activities that disrupt the carbon cycle.
- Destruction of forests and other vegetative cover
 - excessive release of carbon dioxide and other gaseous carbon compounds from industries, vehicles, etc into the atmosphere.
 - bush burning
 - global warming
- (ii) State **two** effects of the disruption of the carbon cycle on the environment. [4 marks]
- depletion of the ozone layer
 - global warming due to the greenhouse effect
 - extinction of certain species of animals
- (d) (i) What is *density* of a body?
- The mass per unit volume of the body

Or

$$\text{Density} = \frac{\text{Mass of body}}{\text{Volume of body}}$$

- (ii) A body of mass 50 kg has a density of 2 kg/m³. Calculate the volume of the body.

[5 marks]

$$\begin{aligned}\text{Volume} &= \frac{\text{Mass}}{\text{Density}} \\ &= \frac{50 \text{ kg}}{2 \text{ kg/m}^3} \\ &= 25 \text{ m}^3\end{aligned}$$

5. (a) What is a *vegetable crop*? [2 marks]

A plant that is grown for food, whose edible parts can be eaten either raw or semi-cooked

- (b) (i) What is air *pollution*?

Contamination of Earth's atmosphere with materials that interfere with human health, the quality of life, or the natural functioning of ecosystems

OR

Introduction into the air of a substance which has harmful or poisonous effects on living things and the earth

- (ii) State **two** human activities that lead to air pollution. [4 marks]

- burning of bush, tyres, etc
- mining
- Driving vehicles that release lots of exhaust fumes
- Construction
- Burning of coal and petroleum in industries

- (c) (i) List **three** modes of heat transfer

- Conduction
- Convection
- Radiation

- (ii) Name the type of medium in which **each** mode of heat transfer you have listed in (i) takes place [6 marks]

- Conduction - solid,
- Convection - liquid, gas
- Radiation - vacuum

- (d) Give **three** reasons why proteins are important in the diet of mammals. [3 marks]

- Builds the body cells / tissues
- Repairs worn out / damaged body cells/tissues

- Help speed up certain chemical reactions in the body
- Help provide energy to the body if carbohydrates and fats are insufficient.

6. (a) State **one** function of **each** of the following parts of the human reproductive system

(i) penis

- Passes semen into the vagina of a woman during sexual intercourse
- Passes urine out of the man's body

(ii) ovary

- Produces ova /eggs (female sex cells)
- Produces reproductive hormones, oestrogen and progesterone

(iii) oviduct

- place where fertilization takes place
- Carries the egg from the ovary to the uterus or womb

(iv) testis

- Produces sperm / spermatozoa (male sex cells)
- Produces the hormone testosterone

(v) uterus

[5 marks]

- place where implantation takes place
- Receives and shelters the embryo during pregnancy

(b) (i) What is *electrical conductor*?

Any material that allows electricity to flow through easily

(ii) List **two** substances which are insulators.

[4 marks]

Plastic, polythene, wood, rubber, paper, fabric, glass, gum

(c) State **three** ways of maintaining soil fertility.

[3 marks]

- Crop rotation
- composting
- mulching
- Fertilizer application
- Bush fallowing
- Green manuring
- Planting of cover crops
- Mixed farming
- crop rotation
- introduction of earthworms and other soil organisms

(d) Write a word equation for **each** of the following chemical reactions:

(iv) oxygen and hydrogen

oxygen + hydrogen → water

or

oxygen + hydrogen → hydrogen peroxide

(v) nitrogen and hydrogen

nitrogen + hydrogen → ammonia

(vi) sodium and chlorine

sodium and chlorine → sodium chloride (common salt)

[3 marks]

END OF PAPER

JUNE 2014

INTEGRATED SCIENCE 1

OBJECTIVE TEST

45 minutes

1. Which of the following types of teeth is used by mammals to tear food material?
 - A. Canines
 - B. Incisors
 - C. Molars
 - D. Premolars

2. The eclipse formed when the moon comes between the sun and the earth is known as
 - A. annular eclipse
 - B. lunar eclipse
 - C. solar eclipse
 - D. total eclipse

3. The element with the chemical symbol **S** is
 - A. silicon
 - B. silver
 - C. sodium
 - D. sulphur

4. The term leaching in soils refers to
 - A. accumulation of organic matter
 - B. decomposition of plant material.
 - C. fixation of nitrogen
 - D. removal of soil nutrients by water.

5. Which of the following statements about a plant cell is **correct**? It
 - A. does not have a nucleus
 - B. contains large vacuoles
 - C. is surrounded by the cell membrane only
 - D. does not have a definite shape

6. An example of a *derived quantity* is
 - A. length
 - B. mass
 - C. temperature
 - D. volume

7. Solid non-metals normally break into pieces when hammered because they are

- A. brittle
 - B. ductile
 - C. lustrous
 - D. malleable
8. Young rabbits are called
- A. bunnies
 - B. cubs
 - C. fingerlings
 - D. kids
9. Producers in an ecosystem are plants that
- A. attract insects
 - B. feed on other plants
 - C. feed on dead materials
 - D. manufacture their own food.
10. The form of energy produced from the nucleus of an atom is termed
- A. chemical energy
 - B. mechanical energy
 - C. nuclear energy
 - D. thermal energy
11. Which of the following elements is a semi-metal?
- A. Carbon
 - B. Nitrogen
 - C. Silicon
 - D. Sodium
12. A system of farming that leads to continuous destruction of virgin forest is
- A. mixed farming
 - B. mixed cropping
 - C. pastoral farming
 - D. shifting farming
13. The by-products of respiration are
- A. carbon dioxide and heat
 - B. carbon dioxide and water
 - C. oxygen and heat
 - D. oxygen and water.
14. Which of the following statements about a force are **correct**? It
- I is measured in newtons
 - II is measured in newton-metre
 - III can start a motion
 - IV can change the direction of a moving body.

- A. I and II only
- B. I and III only
- C. I, III and IV only
- D. I, II, III and IV

15. Which of the following substances is a solid-gas mixture?

- A. Lather
- B. Bronze
- C. Steel
- D. Smoke

16. Weeds on a school farm could be controlled by

- A. handpicking
- B. mowing
- C. ploughing
- D. tilling

17. An example of a non-living tissue used in osmosis experiment is

- A. cellophane
- B. filter paper
- C. potato
- D. polythene

18. Kerosene is poured on the surface of a pond in order to

- A. make it unsafe for human consumption
- B. make it safe for animal consumption
- C. increase the surface tension
- D. break the surface tension

19. Which of the following processes is used to separate insoluble solids from liquids?

- A. Crystallization
- B. Evaporation
- C. Filtration
- D. Sublimation

20. The process of removing unproductive poultry birds from a flock is referred to as

- A. candling
- B. culling
- C. dehorning
- D. drenching

21. Which of the following pairs of diseases can be spread easily when food is exposed to houseflies?

- A. Dysentery and malaria
- B. Malaria and tuberculosis
- C. Dysentery and cholera
- D. Cholera and tuberculosis

22. The pressure in fluids
- A. acts differently in all directions
 - B. acts upwards at any point
 - C. decreases with depth
 - D. increases with depth
23. Which of the following processes involve a change in the state of matter from liquid to solid?
- A. Beans
 - B. Bread
 - C. Cabbage
 - D. Egg
24. Which of the following food items contains the **highest** amount of dietary fibre?
- A. Beans
 - B. Bread
 - C. Cabbage
 - D. Egg
25. An example of a disease vector is
- A. earthworm
 - B. liver fluke
 - C. tapeworm
 - D. tick
26. Which of the following devices is made of semiconductor?
- A. Capacitor
 - B. Inductor
 - C. Resistor
 - D. Transistor
27. Which of the following statements about the properties of water are **correct**? It
- I. is colourless and tasteless
 - II. is neutral to litmus paper
 - III. turns red litmus paper blue
 - IV. is a universal solvent
- A. I and II only
 - B. I and III only
 - C. I, II and IV only
 - D. I, III and IV only
28. The study of the soil profile of an area helps the farmer to
- A. control weed growth
 - B. determine the soil temperature
 - C. determine the types of crop to grow.
 - D. know the pesticides to use.

29. When a green leaf is placed in a test tube containing ethanol and heated over a water bath the leaf
- A. becomes soft
 - B. changes colour to brown
 - C. changes colour to blue-black
 - D. is decolourized.
30. Which of the following sources of energy is / are renewable?
- I. Solar
 - II. Crude oil
 - III. Wind
- A. I only
 - B. I and II only
 - C. I and III only
 - D. I, II and III
31. Which of the following substances is a compound?
- A. Hydrogen
 - B. Nitrogen
 - C. Oxygen
 - D. Water
32. A disease-causing organism that is **most** difficult to control in crop production is
- A. bacterium
 - B. fungus
 - C. nematode
 - D. virus
33. Which of the following organisms is/ are multicellular?
- I. Amoeba
 - II. Paramecium
 - III. Onion
- A. I only
 - B. III only
 - C. I and III only
 - D. II and III only
34. In electronic circuits, LEDs are used to indicate the absence or presence of
- A. emitter and collector
 - B. electric current
 - C. p-n junction
 - D. voltage source.
35. Steel is an alloy of iron and
- A. aluminium
 - B. carbon

- C. silicon
- D. gold

36. Soil aeration could be improved through

- A. mulching
- B. irrigation
- C. soil drainage
- D. fertilizer application

37. Which of the following pairs of structures are part of the respiratory system of humans?

- A. Fallopian tube and alveoli
- B. Pharynx and oesophagus
- C. Trachea and alveoli
- D. Trachea and duodenum

38. Which of the following effects is **not** a result of illegal connection of electricity? It can

- A. lead to fire outbreak
- B. lead to frequent power cut-off
- C. increase the flow of current in the supply chain.
- D. cause damage to electrical appliance

39. Which of the following statements about the scientific method is / are **correct**? It provides

- I. logical procedure for arriving at knowledge.
- II. knowledge that can be verified
- III. knowledge that can never be changed

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only

40. An entrepreneur's decision as to what to produce is a/ an

- A. co-ordinating function
- B. organizing function
- C. planning function
- D. supervising function.

END OF PAPER

JUNE 2014

INTEGRATED SCIENCE 1

OBJECTIVE TEST

SOLUTIONS

1. A. canines
2. C. solar eclipse
3. D. sulphur
4. D. removal of soil nutrients by water
5. B. contains large vacuoles
6. D. volume
7. A. brittle
8. A. bunnies
9. D. manufacture their own food
10. C. nuclear energy
11. C. silicon
12. D. shifting farming
13. B. carbon dioxide and water
14. C. I, III and IV only
15. D. smoke
16. A. handpicking
17. A. cellophane
18. D. break the surface tension
19. C. filtration
20. B. culling
21. C. Dysentery and cholera
22. D. increases with depth
23. B. freezing
24. A. Beans
25. D. tick
26. D. Transistor
27. C. I, II and IV only
28. C. determine the types of crop to grow

- 29. D. decolourized
- 30. C. I and III only
- 31. D. water
- 32. A. Bacterium
- 33. B. III only
- 34. B. electric current
- 35. B. carbon
- 36. D. fertilizer application
- 37. C. trachea and alveoli
- 38. C. increase the flow of current in the supply chain
- 39. B. I and II only
- 40. C. planning function

JUNE 2014

INTEGRATED SCIENCE 2

ESSAY

1 ¼ hours

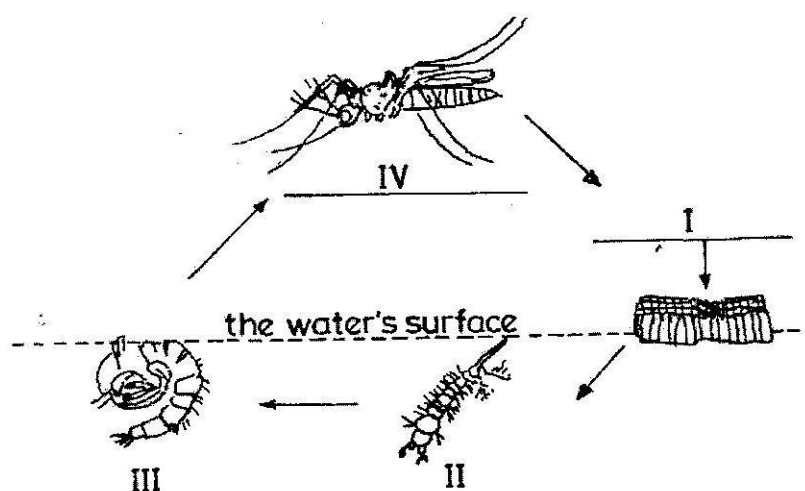
PART I

[40 marks]

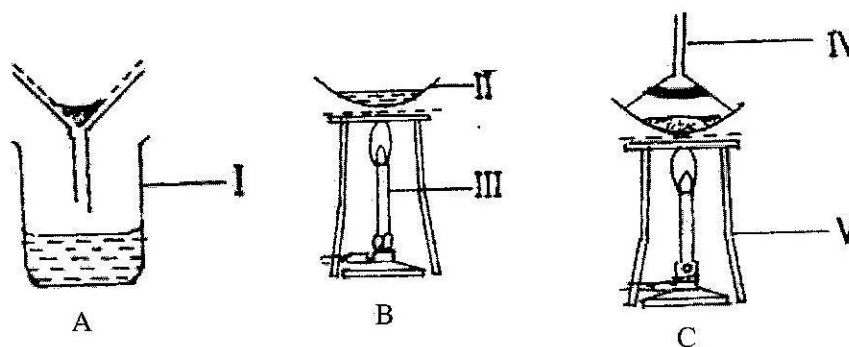
Answer **all** of Question 1

1. (a) The diagrams below represent the stages in the life cycle of a mosquito

Study the diagrams carefully and answer the questions that follow

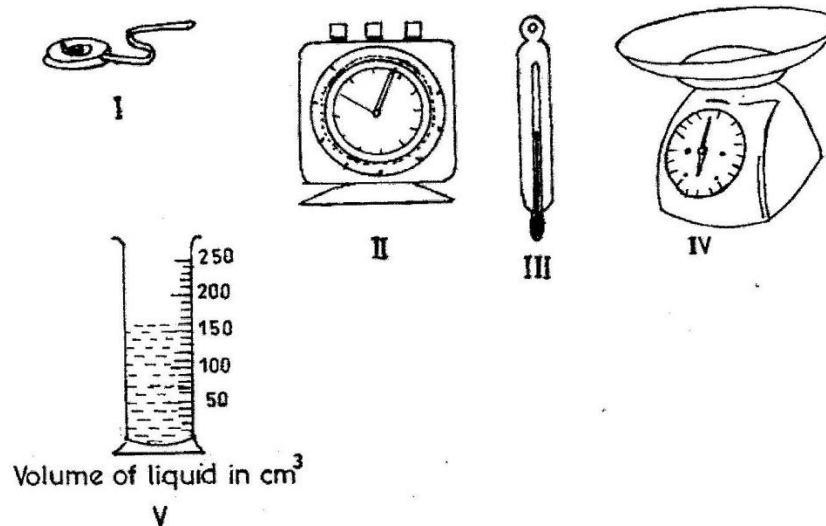


- (i) Name **each** of the stages labeled **I, II, III** and **IV**
(ii) State how stage **II** obtains oxygen
(iii) State **two** methods of controlling **each** of the stages labeled **III** and **IV**
- (b) The diagrams below are different laboratory set-ups used in the separation of mixtures. Study the diagrams carefully and answer the questions that follow

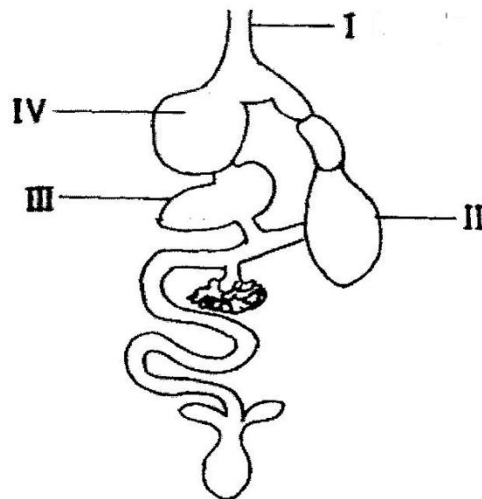


- (i) Name **each** of the parts labeled **I, II, III, IV** and **V**.

- (ii) Name the separation method represented by **each** diagram.
 - (iii) Which of the set-ups is used to obtain clear water from muddy water?
 - (iv) Which of the set-ups is used to obtain salt from salt solution?
- (c) The diagrams below show some instruments used in the laboratory.
Study the diagrams carefully and answer the questions that follow



- (i) Identify **each** of the instruments labeled **I, II, III, IV** and **V**
 - (ii) State **one** use of **each** of the instruments labeled **I, II, III** and **IV**
 - (iii) Read and record the volume of the liquid in the instrument labeled **V**
- (d) The diagram below shows the digestive system of a class of farm animals.
Study the diagrams carefully and answer the questions that follow



- (i) Name **each** of the parts labeled **I, II, III** and **IV**
- (ii) State **one** function **each** of the parts labeled **II** and **IV**
- (iii) Name **two** farm animals that possess this type of digestive system.
- (iv) Mention **two** diseases which affect this class of farm animals.

[10 marks]

PART II
[60 marks]

Answer **four** questions only from this part.

2. (a) (i) Name the **two** elements that combine to form water.
(ii) Write a balanced chemical equation to show how the water is formed from the named elements
- (b) State **two** ways of maintaining a balance in an ecosystem.
- (c) (i) What is a *fertile soil*?
(ii) State **two** factors that cause loss of soil fertility.
- (d) Classify the following items as *magnetic* or *non-magnetic* substance:
wood, steel blade, rubber and glass jar.
3. (a) (i) What is *germination of seed*?
(ii) State **two** conditions necessary for the germination of seed.
- (b) State **four** methods used in identifying farm animals
- (c) Explain why it is easier to cut a piece of yam with a sharp knife than with a blunt knife
- (d) State **three** differences between a *metal* and a *non-metal*.
4. (a) (i) What is *debeaking*?
(ii) Give two reasons why debeaking in poultry birds is important.
- (b) (i) A steel needle carefully placed on the surface of water floats. What type of force made the steel needle to float?
(ii) Name **three** substances that could be added to the water to make the steel needle to sink.
- (c) (i) Explain why gold is preferred to iron in the making of jewelleryes.
(ii) State **one** way of preventing rusting.
- (d) (i) State **two** elements of climate
(ii) Name the equipment used to measure **each** of the elements stated in (i) above.
5. (a) (i) What is *refraction of light*?
(ii) Sketch a diagram to show the path of a light ray when it travels from air to glass.

- (b) Explain why it is difficult to separate iron and sulphur mixture after strong heating.
- (c)
 - (i) Give **two** example of digestive enzymes produced in humans.
 - (ii) For **each** of the enzymes given in (i), name the part of the human body where the enzyme is produced.
- (d) List **four** methods of applying fertilizers to crops.

6. (a) Consider the substance listed below:
carbon dioxide, gold, bronze, iron, oxygen and ink

From the list, select the substance that:

- (i) supports burning
 - (ii) is used as jewellery;
 - (iii) is used for making statues
- (b)
 - (i) Name **two** diseases associated with the circulatory system of humans.
 - (ii) State **one** way of preventing **each** of the diseases named in (i)
- (c) Give two examples of **each** of:
 - (i) **major** plant nutrients;
 - (ii) **minor** plant nutrients.
- (d)
 - (i) State **two** properties of a good thermometric liquid.
 - (ii) Give **two** examples of a good thermometric liquid.

END OF ESSAY TEST

JUNE 2014

INTEGRATED SCIENCE 2

ESSAY

SOLUTIONS

1. (a)
- (i) Name of each stage
- I. Egg
 - II. Larva
 - III. Pupa
 - IV. Adult / Imago
- (ii) How stage **II** obtains oxygen
- The larva comes to the surface of the water body to obtain oxygen from the air through a structure called the siphon.
- (iii) Methods of controlling **each** of the stages labeled **III** and **IV**
- Controlling stage III**
- Adding oil to cover the water surface
 - Introduction of fish into the water body
 - Spraying the water body with pesticides
 - Adding kerosene to cover the water surface
- Controlling stage IV**
- The use of lethal ovitraps
 - The use of mosquito spray / insecticide
 - The use of mosquito coil
 - Clearing mosquito breeding grounds such as choked gutters, stagnant pools of water, etc
- (b)
- (i) Name of the part labeled
- I. Beaker
 - II. Evaporating disc
 - III. Candle
 - IV. Inverted funnel
 - V. Tripod stand
- (ii) Name the separation method represented by **each** diagram.
- A- Filtration

B- Evaporation

C- sublimation

(iii) The set-up used to obtain clear water from muddy water

Set up A

(iv) The set-up used to obtain salt from salt solution?

Set-up B

(c)

(i) Identification of the instruments labeled **I, II, III, IV** and **V**

I. Tape measure

II. stop clock

III. Thermometer

IV. Weighing scale/ Top pan balance

V. Measuring cylinder

(ii) State **one** use of **each** of the instruments labeled **I, II, III** and **IV**

I. to measure length of materials

II. to measure time

III. to measure temperature

IV. to measure the mass of materials

V. to measure the volumes of liquid substances

(iii) Read and record the volume of the liquid in the instrument labeled **V**

Volume of liquid = 160 cm³

(d)

(i) Name **each** of the parts labeled **I, II, III** and **IV**

I. - Gullet

II. - Gizzard

III. - Liver

IV. - Crop

(ii) State **one** function **each** of the parts labeled **II** and **IV**

I. - used to grind food

II. - used to store food temporarily

(iii) Name **two** farm animals that possess this type of digestive system.

Turkey, fowl, duck, ostrich, geese, guinea fowl.

(iv) Mention **two** diseases which affect this class of farm animals.

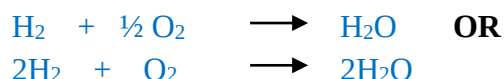
Coccidiosis, fowl typhoid, fowl pox, newcastle disease, pneumonia.

PART II
[60 marks]

2. (a) (i) Name the **two** elements that combine to form water.

Hydrogen and oxygen

- (ii) Write a balanced chemical equation to show how the water is formed from the named elements



- (b) State **two** ways of maintaining a balance in an ecosystem.

- Stopping indiscriminate felling of trees
- By enforcing environmental protection laws; United Nations resolutions on release of poisonous gases should be adhered to.
- Activities of mining industries should be strictly monitored.
- Poaching of animals for their parts should be stopped

- (c) (i) What is a *fertile soil*?

Soil that has the ability to supply the required nutrients needed for plant growth

- (ii) State **two** factors that cause loss of soil fertility.

- Soil erosion,
- leaching,
- excessive irrigation,
- over cropping,
- surface compacting.

- (d) Classify the following items as *magnetic* or *non-magnetic* substance:
wood, steel blade, rubber and glass jar.

Magnetic substance - steel blade

Non-magnetic substance - wood, rubber and glass jar.

3. (a) (i) What is *germination of seed*?

The process by which a viable seed grows/develops into a seedling.

- (ii) State **two** conditions necessary for the germination of seed.

- Presence of air
- Presence of water
- Viable seed

- Optimum temperature
- (b) State **four** methods used in identifying farm animals
Tagging, tattooing, branding, tonging, ear notching
- (c) Explain why it is easier to cut a piece of yam with a sharp knife than with a blunt knife
The cutting edge of a sharp knife has very small surface area so requires smaller force to yield the pressure needed to cut the yam - making cutting easy, but the cutting edge of a blunt knife has a relatively larger surface area so it needs a larger force to yield the pressure needed to cut the yam.
- (d) State **three** differences between a *metal* and a *non-metal*.

Metals	Non-metals
Have high melting point	Have low melting point
Are lustrous	Are not lustrous
Are malleable	Are not malleable
Have high density	Have low density
Are ductile	Are brittle
Are good conductors of heat and electric current	Are poor conductors of heat and electric current

- 4.
- (a) (i) What is *debeaking*?
The process by which about half the beak of a bird is removed
or
The process of removing the upper part of the beak of a bird
- (ii) Give **two** reasons why debeaking in poultry birds is important.
- To prevent wasting of food
 - To prevent egg eating
 - To prevent vent pecking
 - To prevent injuring other birds
- (b) (i) A steel needle carefully placed on the surface of water floats. What type of force made the steel needle to float?
Surface tension
- (ii) Name **three** substances that could be added to the water to make the steel needle to sink.
Oil, detergent, grease, kerosene, soap
- (c) (i) Explain why gold is preferred to iron in the making of jewelleryes.
It does not react with substances like water and oxygen.
It does not rust
It is very malleable

- (ii) State **one** way of preventing rusting.
- Applying oil to the surface of the metal
 - Galvanizing the metal
 - Keeping the metal inside a desiccator
 - Alloying the metal

- (d) (i) State **two** elements of climate

Temperature,
rainfall,
relative humidity,
light intensity,
wind speed,
wind direction,
atmospheric pressure

- (ii) Name the equipment used to measure **each** of the elements stated in (i) above.

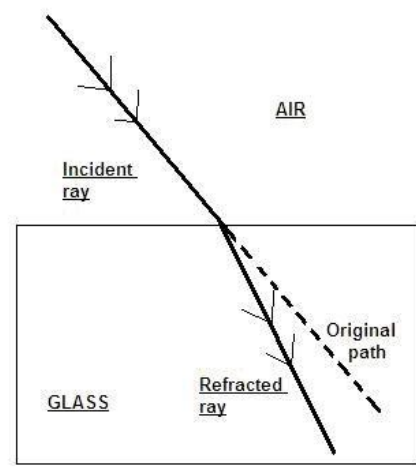
<u>ELEMENT</u>	<u>EQUIPMENT</u>
Temperature	Thermometer
rainfall,	rain gauge
relative humidity,	hygrometer
light intensity,	photometer
wind speed,	wind vane
wind direction,	anemometer
atmospheric pressure	barometer

5.

- (a) (i) What is *refraction of light*?

The bending of light rays as it travels from an optically less dense medium to an optically more dense medium

- (ii) Sketch a diagram to show the path of a light ray when it travels from air to glass.



- (b) Explain why it is difficult to separate iron and sulphur mixture after strong heating.

This is because after strong heating, the iron and sulphur react to form iron (II) sulphide which cannot be separated by physical means

- (c) (i) Give **two** example of digestive enzymes produced in humans.

Salivary amylase,
pepsin,
renin,
lipase,
trypsin,
pancreatic amylase,
maltase

- (ii) For **each** of the enzymes given in (i), name the part of the human body where the enzyme is produced.

<u>ENZYME</u>	<u>PART WHERE ENZYME IS PRODUCED</u>
Salivary amylase	Mouth
pepsin,	stomach
rennin	stomach
lipase	pancreas
trypsin	pancreas
pancreatic amylase	pancreas
maltase	small intestine

- (d) List **four** methods of applying fertilizers to crops.

Drilling, broadcasting, side dressing , top dressing, ringing,

6.

- (a) Consider the substance listed below:

carbon dioxide, gold, bronze, iron, oxygen and ink

From the list, select the substance that:

- (i) supports burning - oxygen
(ii) is used as jewellery; - gold
(iii) is used for making statues - bronze

- (b) (i) Name **two** diseases associated with the circulatory system of humans.

Arteriosclerosis, coronary thrombosis, heart cancer, leukimia, high blood pressure, piles,
varicose vein

- (ii) State **one** way of preventing **each** of the diseases named in (i)

Arteriosclerosis-

- regular exercising of the body
- avoid smoking
- low fat intake

Piles-

- intake a lot of water
- eat a lot of vegetables and fruits

High blood pressure-

- regular exercising of the body
- low salt intake
- low fat intake

Varicose vein- surgical removal of vein

- (c) Give two examples of **each** of:
- (i) **major** plant nutrients;
Nitrogen, phosphorus, sulphur, potassium, calcium, magnesium.
- (ii) **minor** plant nutrients.
Chlorine, iron, fluoride, manganese, molybdenum
- (d) (i) State **two** properties of a good thermometric liquid.
- Remains a liquid within a wide temperature range
 - It must not wet glass
 - Expands uniformly with a rise in temperature
 - It must not vaporize easily
- (ii) Give **two** examples of a good thermometric liquid.
Mercury and alcohol

END OF ESSAY TEST